# DATA VISUALISATION & FIGURE DESIGN

FCP

Matt Castle

# EXERCISE

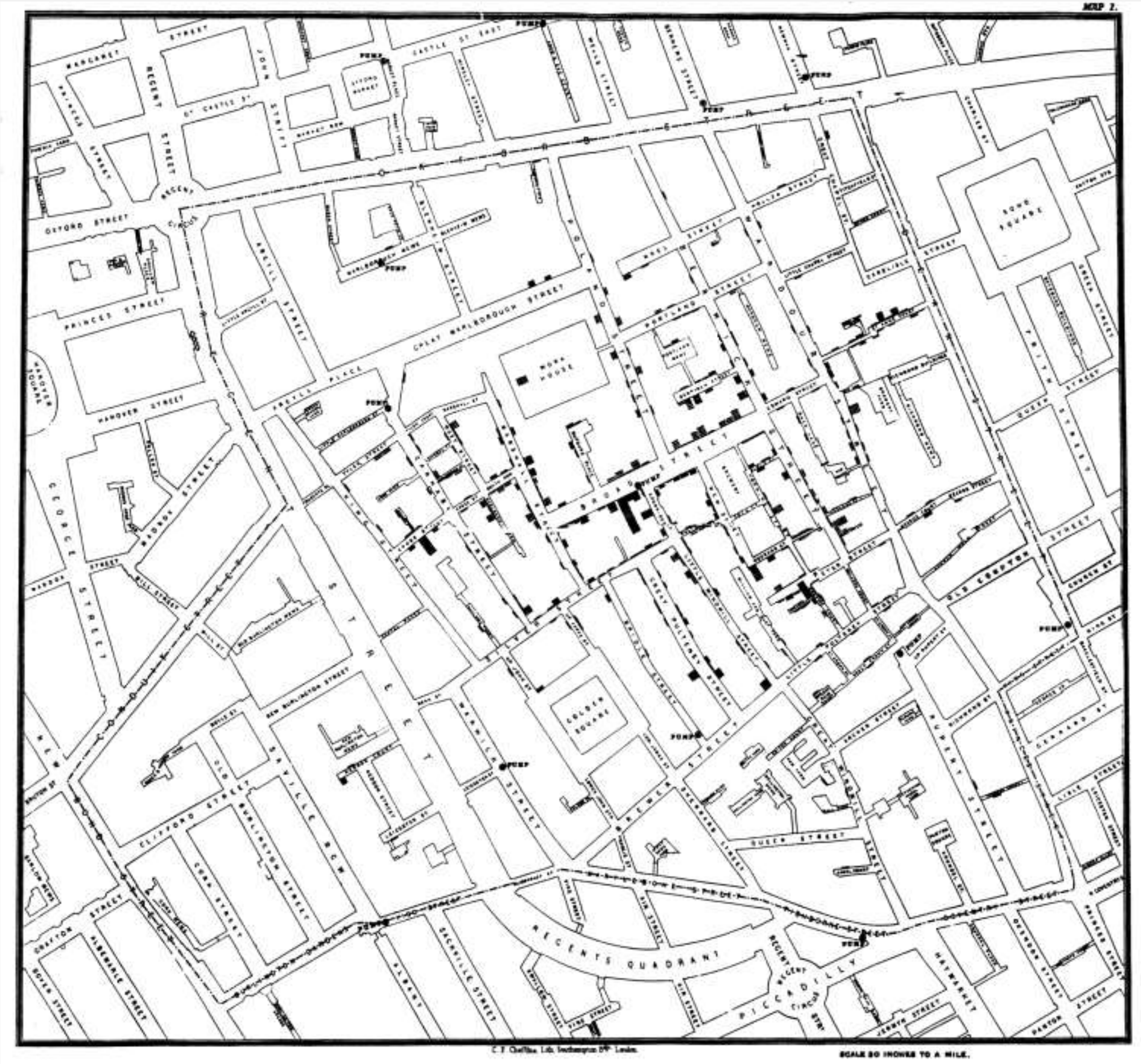
- Split into groups of 4
- Find a visualisation online to share with the group
- Send the link to my email (ajv37@cam.ac.uk)

# TODAY'S SESSION

- Purpose
- Design Approaches
- Visualisation Concepts
- Group Exercise

# WHAT IS THE PURPOSE OF DATA VISUALISATION?

| Easting | Northing | Deaths |
|---|---|---|
| 529,308.7 | 181,031.4 | 3 |
| 529,312.2 | 181,025.2 | 2 |
| 529,314.4 | 181,020.3 | 1 |
| 529,317.4 | 181,014.3 | 1 |
| 529,320.7 | 181,007.9 | 4 |
| 529,336.7 | 181,006.0 | 2 |
| 529,290.1 | 181,024.4 | 2 |
| 529,301.0 | 181,021.2 | 2 |
| 529,285.0 | 181,020.2 | 3 |
| 529,288.4 | 181,031.8 | 2 |
| … | … | … |


MAP 1.
OXFORD STREET
REGENT STREET
BROAD STREET
GOLDEN SQUARE
SOHO SQUARE
WORK HOUSE
PUMP
REGENTS QUADRANT
PICCADILLY
SCALE 30 INCHES TO A MILE.

| Description | Amount ($Billion) |
|---|---|
| Global pharma market | 950 |
| US budget deficit | 901 |
| China | 90 |
| Tobacco | 740 |
| US | 739 |
| Carlos Slim Helu | 73 |
| Illegal drugs market | 672 |
| Bill Gates | 67 |
| Video games | 67 |
| Eurozone rescue fund | 650 |
| Movies | 64 |
| … | … |

# The Billion Dollar-o-Gram


David McCandless // v2.0 // Jul 2010
research: James Key, Caroline Flyn, Matthew Sawh
InformationIsBeautiful.net

sources: media reports (NYtimes, Guardian, BBC)
numbers rounded, slight visual cheating to make things fit
data: **bit.ly/billions**

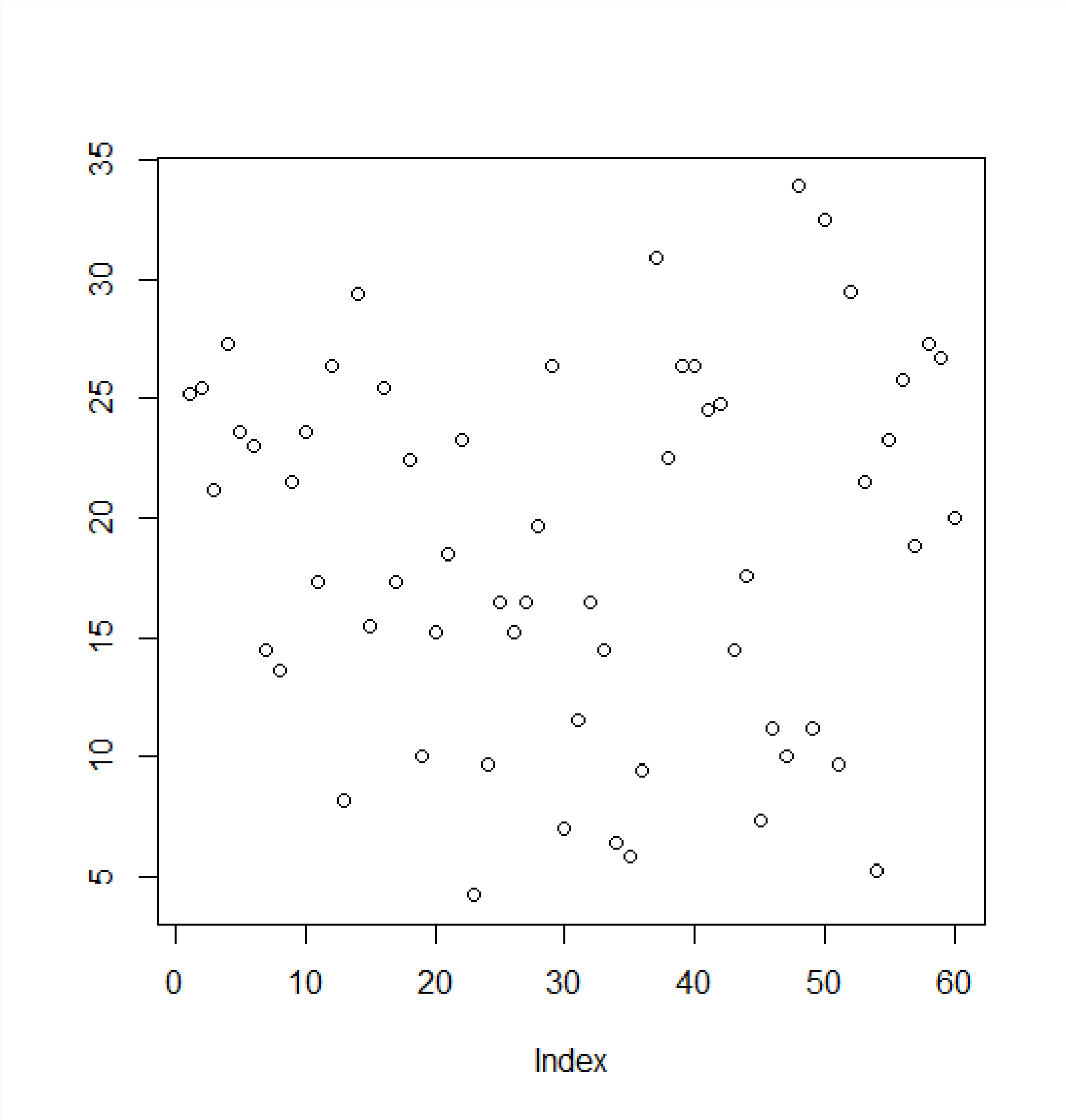
# DATA VISUALISATION

- Two aims:
  - Explore data to understand its key features and discover patterns
  - Communicate key features of data clearly (Tell a story)

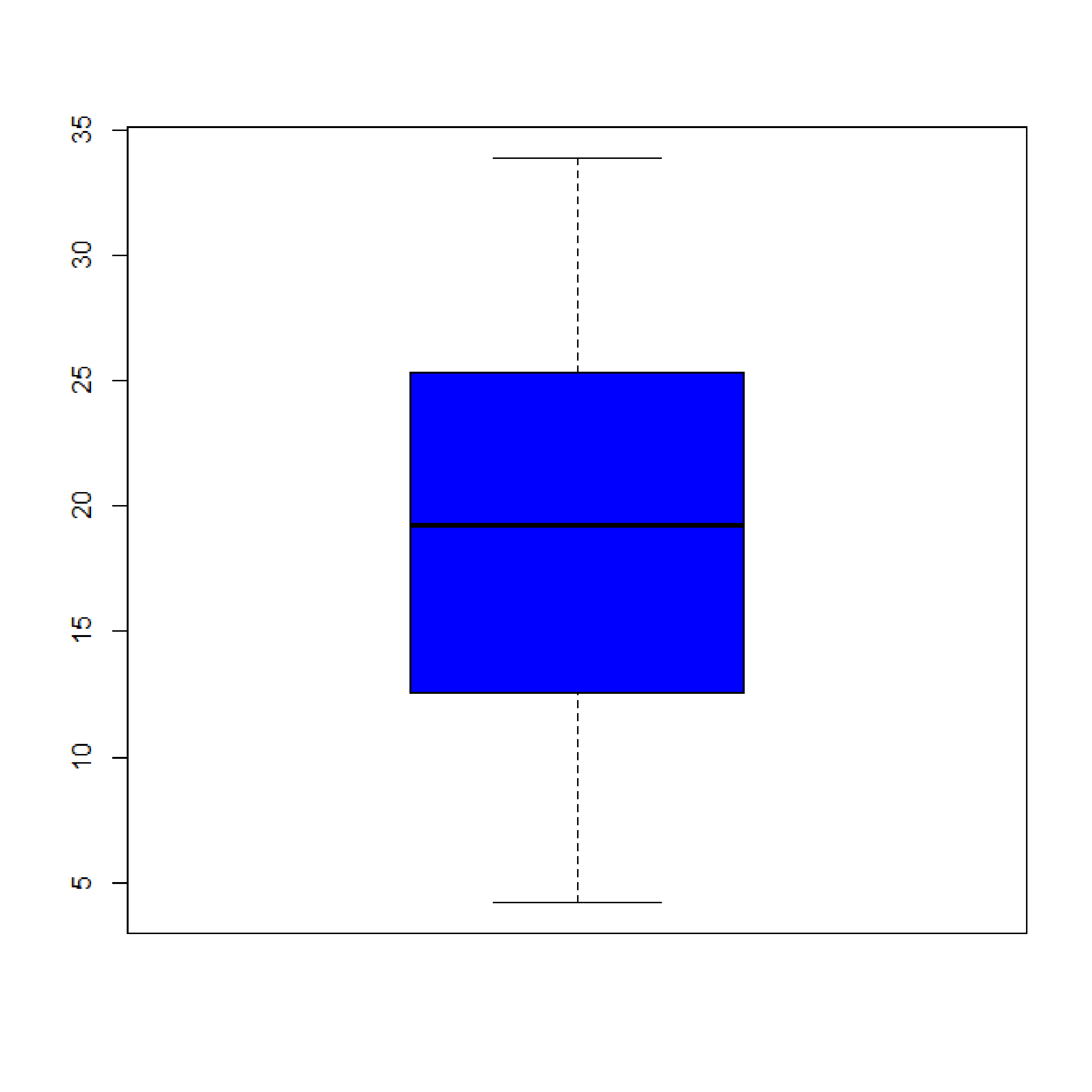
# EXPLORATION

| | | | | | |
|---|---|---|---|---|---|
| 4.2 | 16.5 | 23.6 | 15.2 | 19.7 | 25.5 |
| 11.5 | 16.5 | 18.5 | 21.5 | 23.3 | 26.4 |
| 7.3 | 15.2 | 33.9 | 17.6 | 23.6 | 22.4 |
| 5.8 | 17.3 | 25.5 | 9.7 | 26.4 | 24.5 |
| 6.4 | 22.5 | 26.4 | 14.5 | 20 | 24.8 |
| 10 | 17.3 | 32.5 | 10 | 25.2 | 30.9 |
| 11.2 | 13.6 | 26.7 | 8.2 | 25.8 | 26.4 |
| 11.2 | 14.5 | 21.5 | 9.4 | 21.2 | 27.3 |
| 5.2 | 18.8 | 23.3 | 16.5 | 14.5 | 29.4 |
| 7 | 15.5 | 29.5 | 9.7 | 27.3 | 23 |

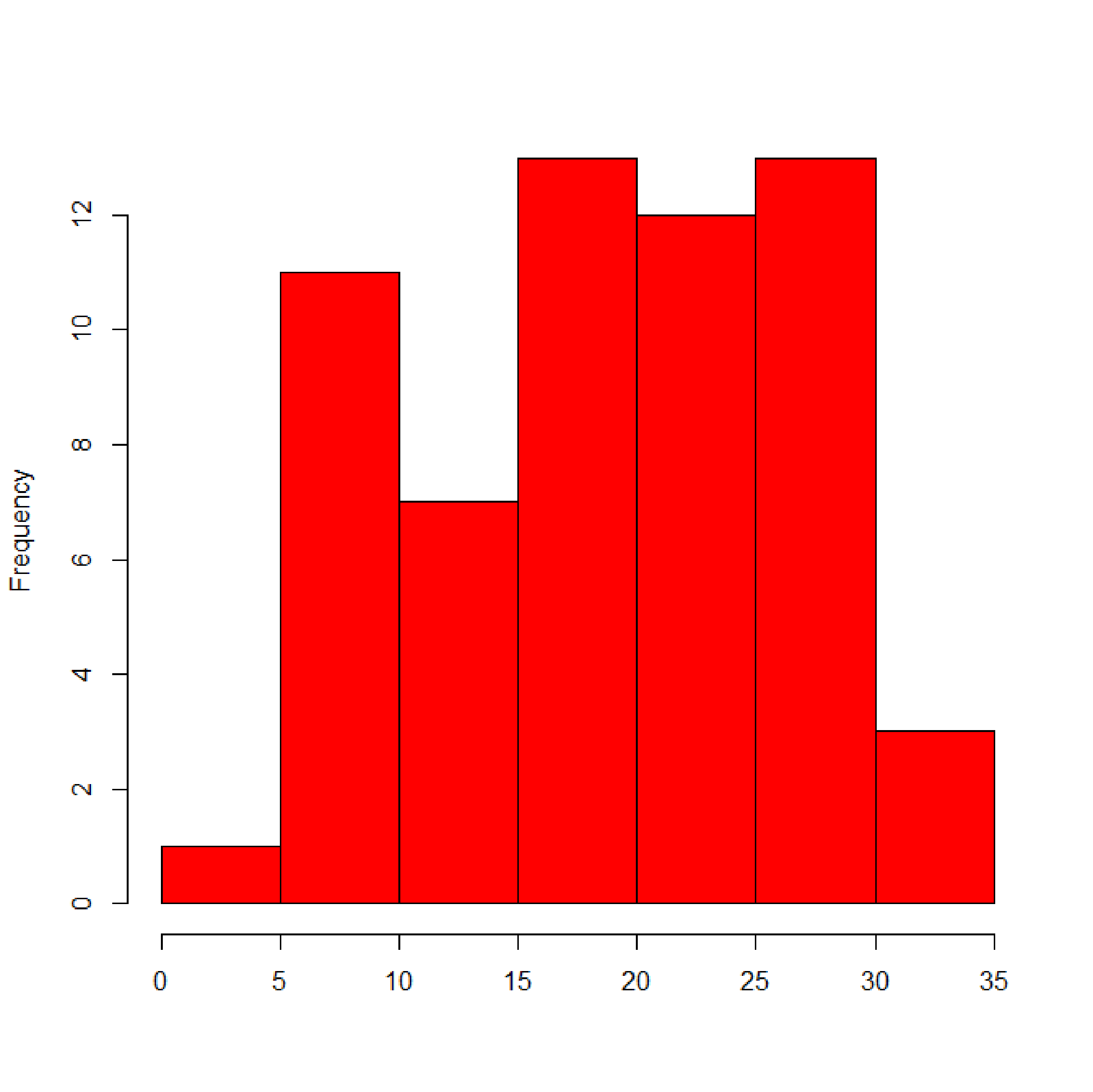
Raw Data



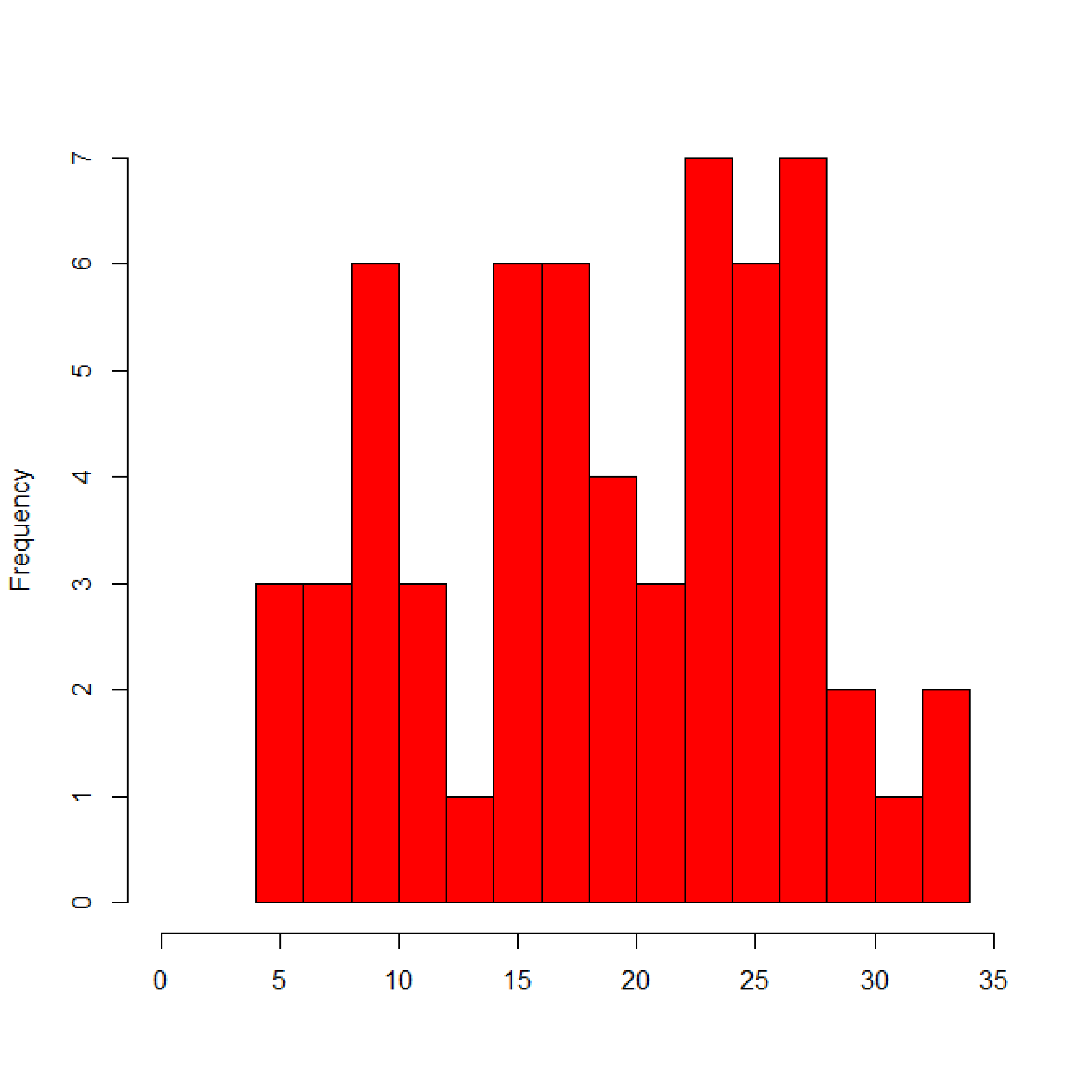

Scatterplot of value against index



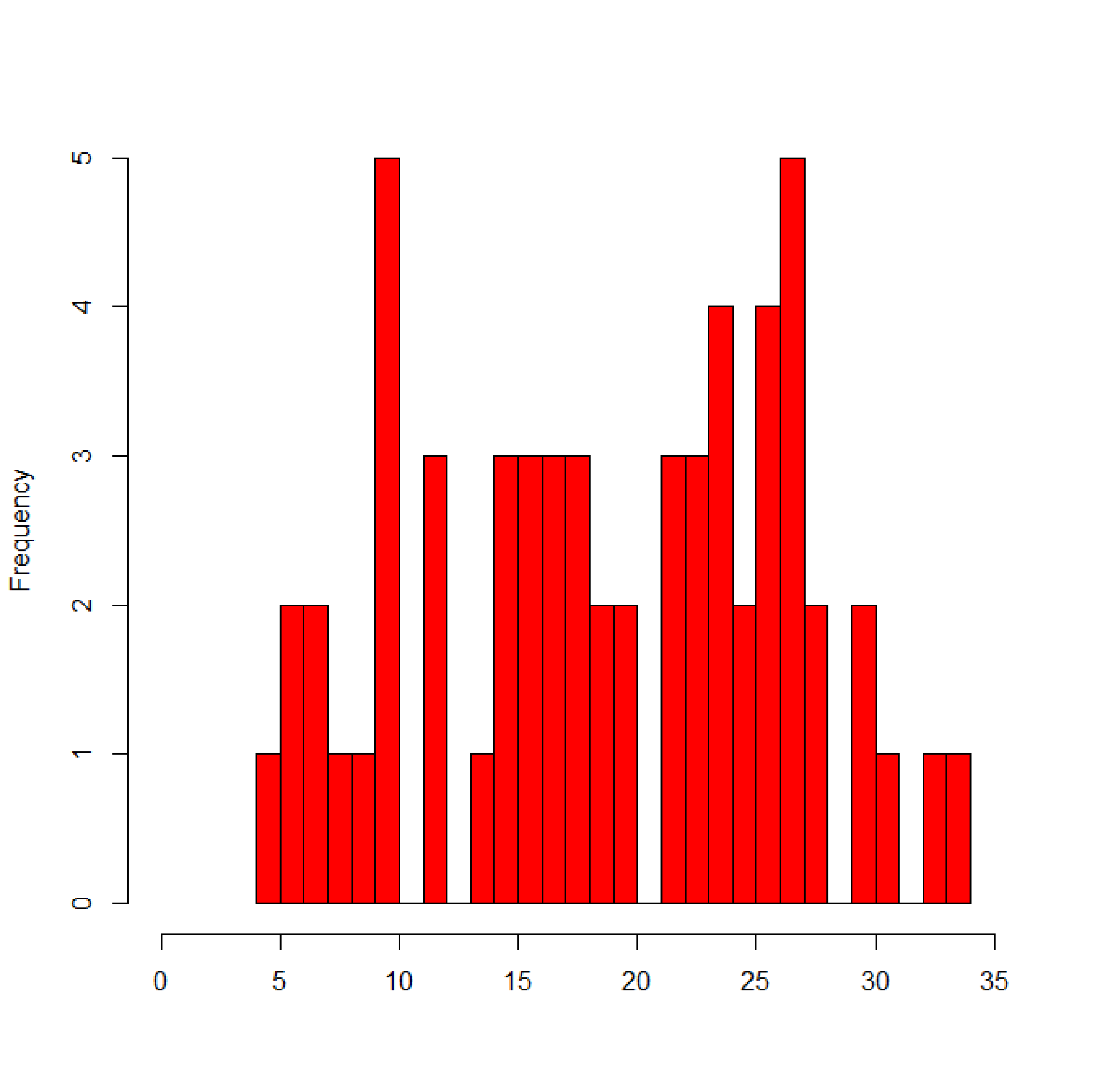

Boxplot




Histogram with 7 bins

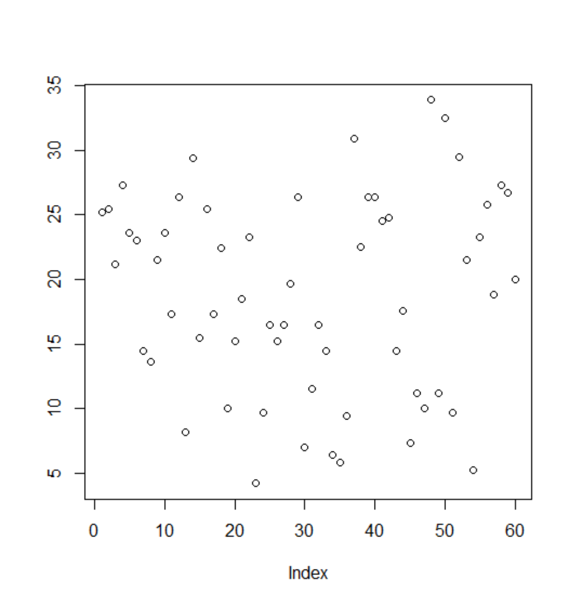


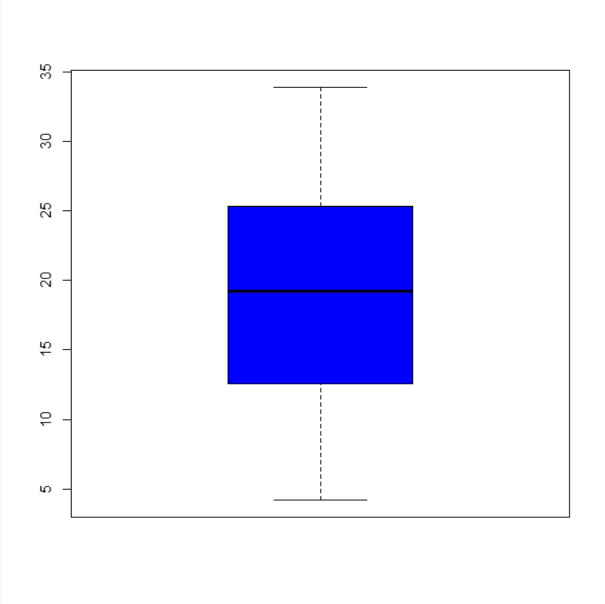

Histogram with 15 bins

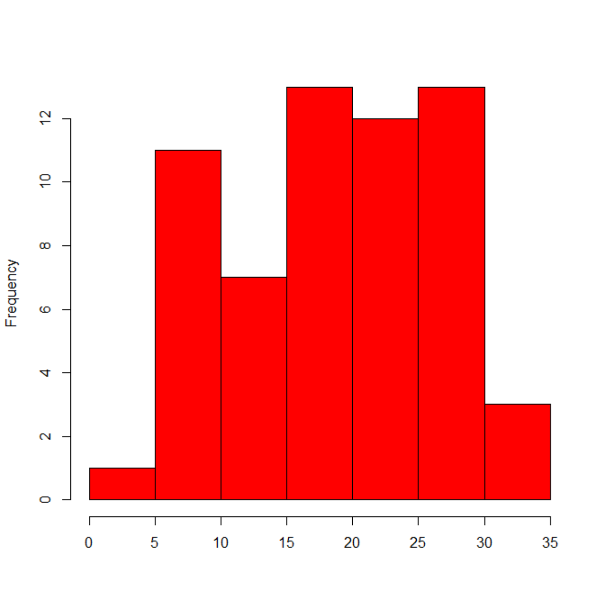


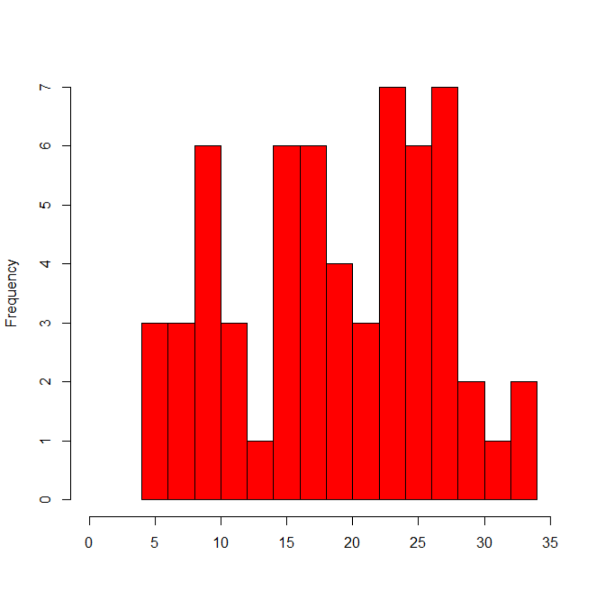

Histogram with 30 bins

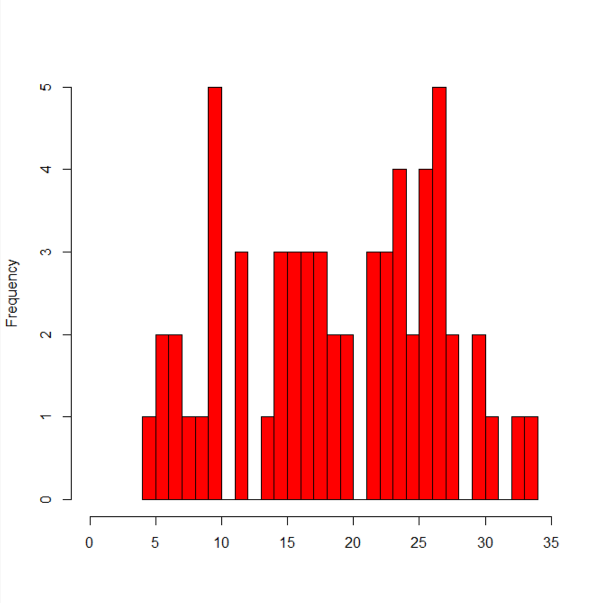
| | | | | | |
|---|---|---|---|---|---|
| 4.2 | 16.5 | 23.6 | 15.2 | 19.7 | 25.5 |
| 11.5 | 16.5 | 18.5 | 21.5 | 23.3 | 26.4 |
| 7.3 | 15.2 | 33.9 | 17.6 | 23.6 | 22.4 |
| 5.8 | 17.3 | 25.5 | 9.7 | 26.4 | 24.5 |
| 6.4 | 22.5 | 26.4 | 14.5 | 20 | 24.8 |
| 10 | 17.3 | 32.5 | 10 | 25.2 | 30.9 |
| 11.2 | 13.6 | 26.7 | 8.2 | 25.8 | 26.4 |
| 11.2 | 14.5 | 21.5 | 9.4 | 21.2 | 27.3 |
| 5.2 | 18.8 | 23.3 | 16.5 | 14.5 | 29.4 |
| 7 | 15.5 | 29.5 | 9.7 | 27.3 | 23 |



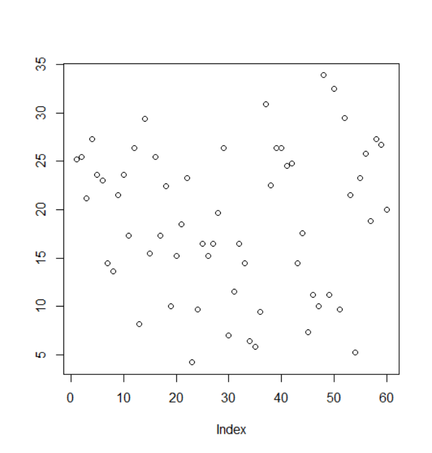
# EXPLORATION SUMMARY

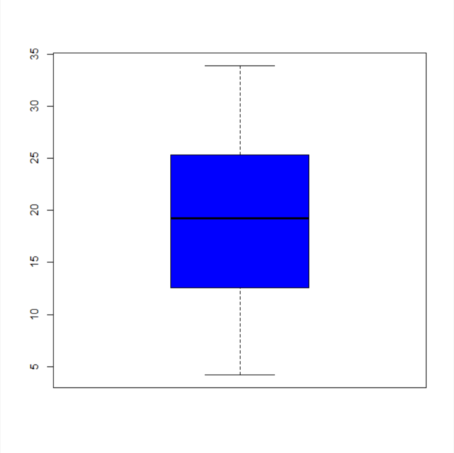
- Aim is to determine patterns in the data
- Get information about:
  - Trends
  - Distributions
  - Correlations / comparisons between groups

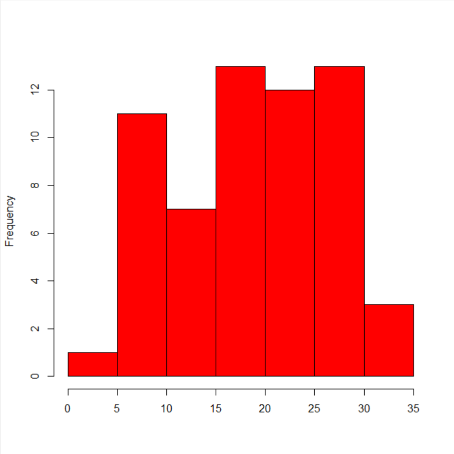
More visualisations means more patterns

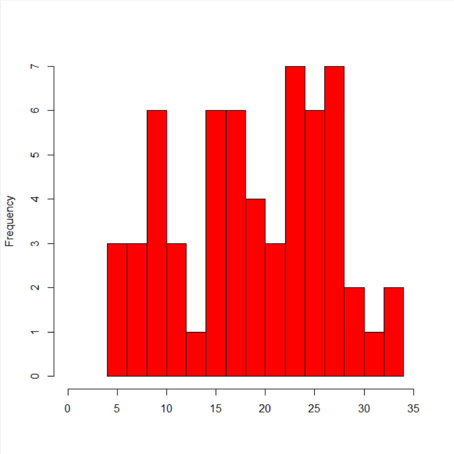
# COMMUNICATION

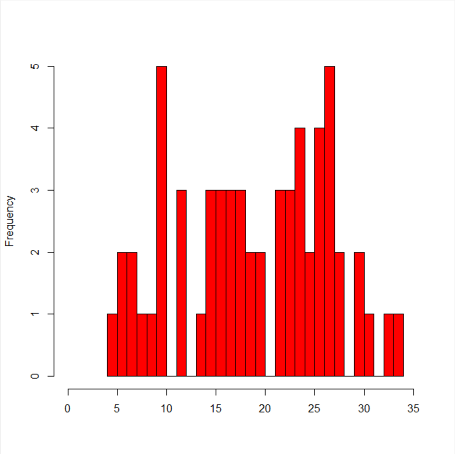
| | | | | | |
|---|---|---|---|---|---|
| 4.2 | 16.5 | 23.6 | 15.2 | 19.7 | 25.5 |
| 11.5 | 16.5 | 18.5 | 21.5 | 23.3 | 26.4 |
| 7.3 | 15.2 | 33.9 | 17.6 | 23.6 | 22.4 |
| 5.8 | 17.3 | 25.5 | 9.7 | 26.4 | 24.5 |
| 6.4 | 22.5 | 26.4 | 14.5 | 20 | 24.8 |
| 10 | 17.3 | 32.5 | 10 | 25.2 | 30.9 |
| 11.2 | 13.6 | 26.7 | 8.2 | 25.8 | 26.4 |
| 11.2 | 14.5 | 21.5 | 9.4 | 21.2 | 27.3 |
| 5.2 | 18.8 | 23.3 | 16.5 | 14.5 | 29.4 |
| 7 | 15.5 | 29.5 | 9.7 | 27.3 | 23 |





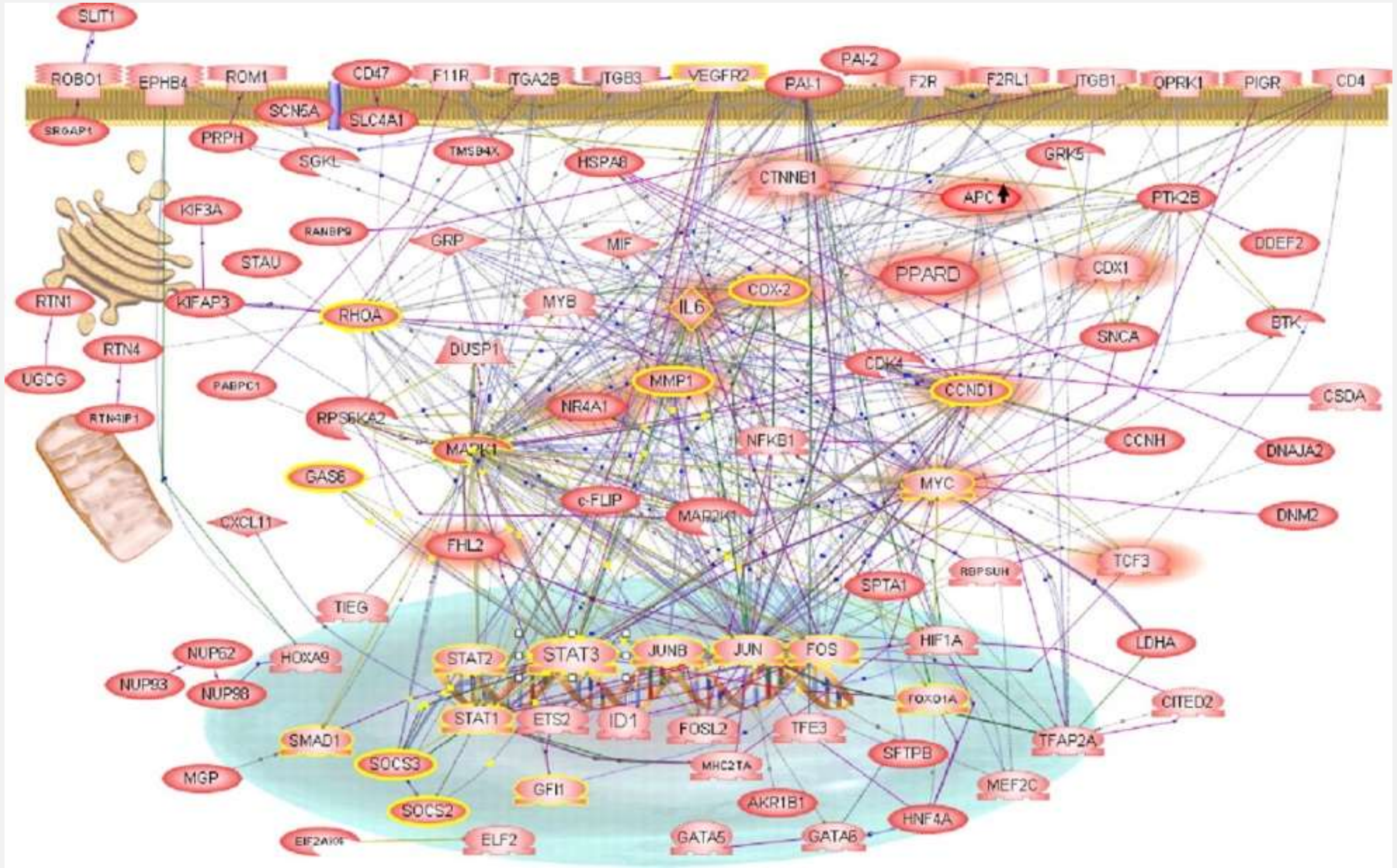
# COMMUNICATION

1. Consider the message you want to get across
2. Consider the audience
3. Acknowledge the strengths and weaknesses to different representations
4. There isn't a single **correct** representation

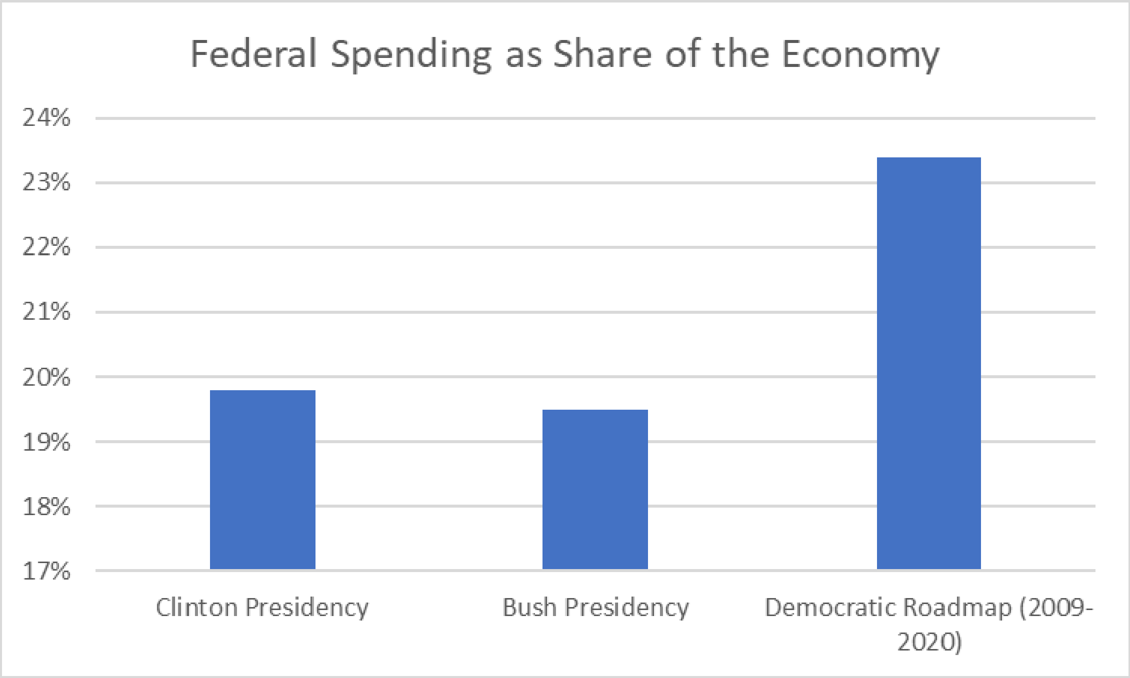
There is always a **story** to tell

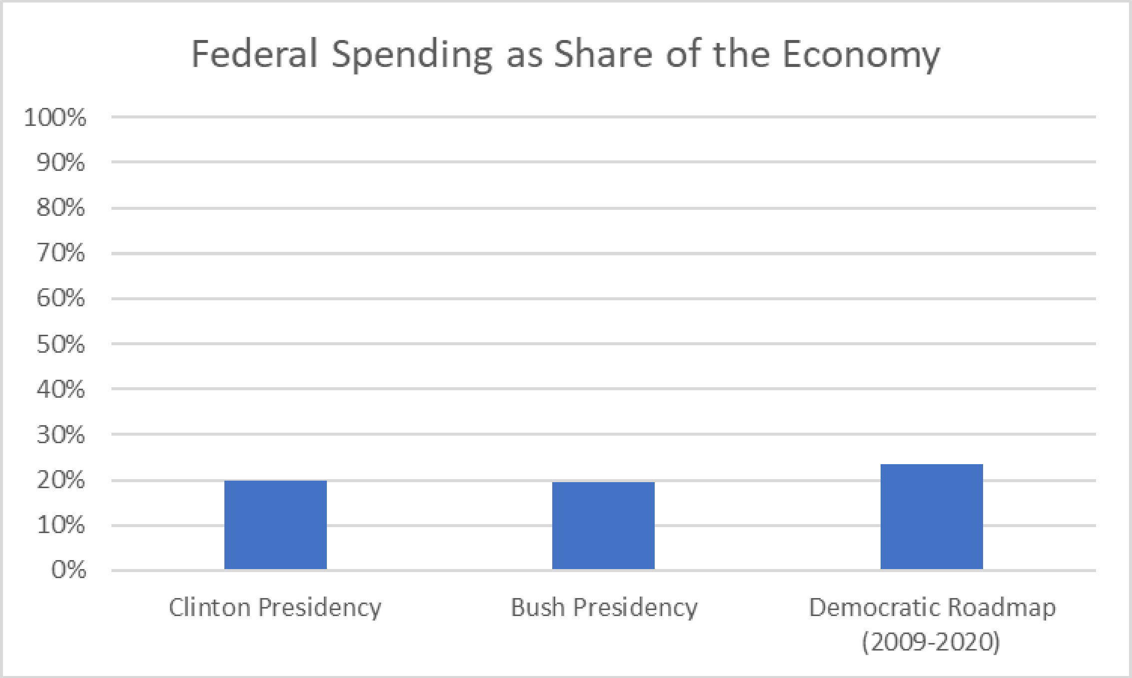
# POOR COMMUNICATION

No story
No clarity

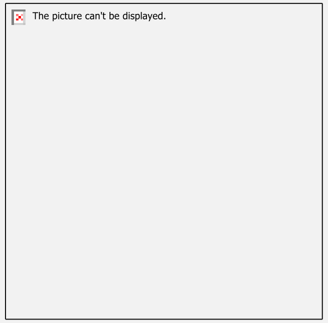


# COMMUNICATION DECISIONS



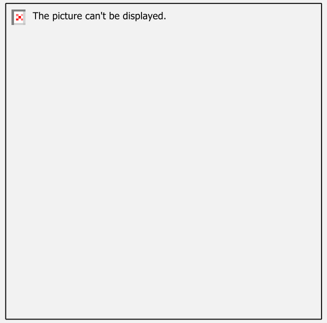



# DATA VISUALISATION PURPOSES



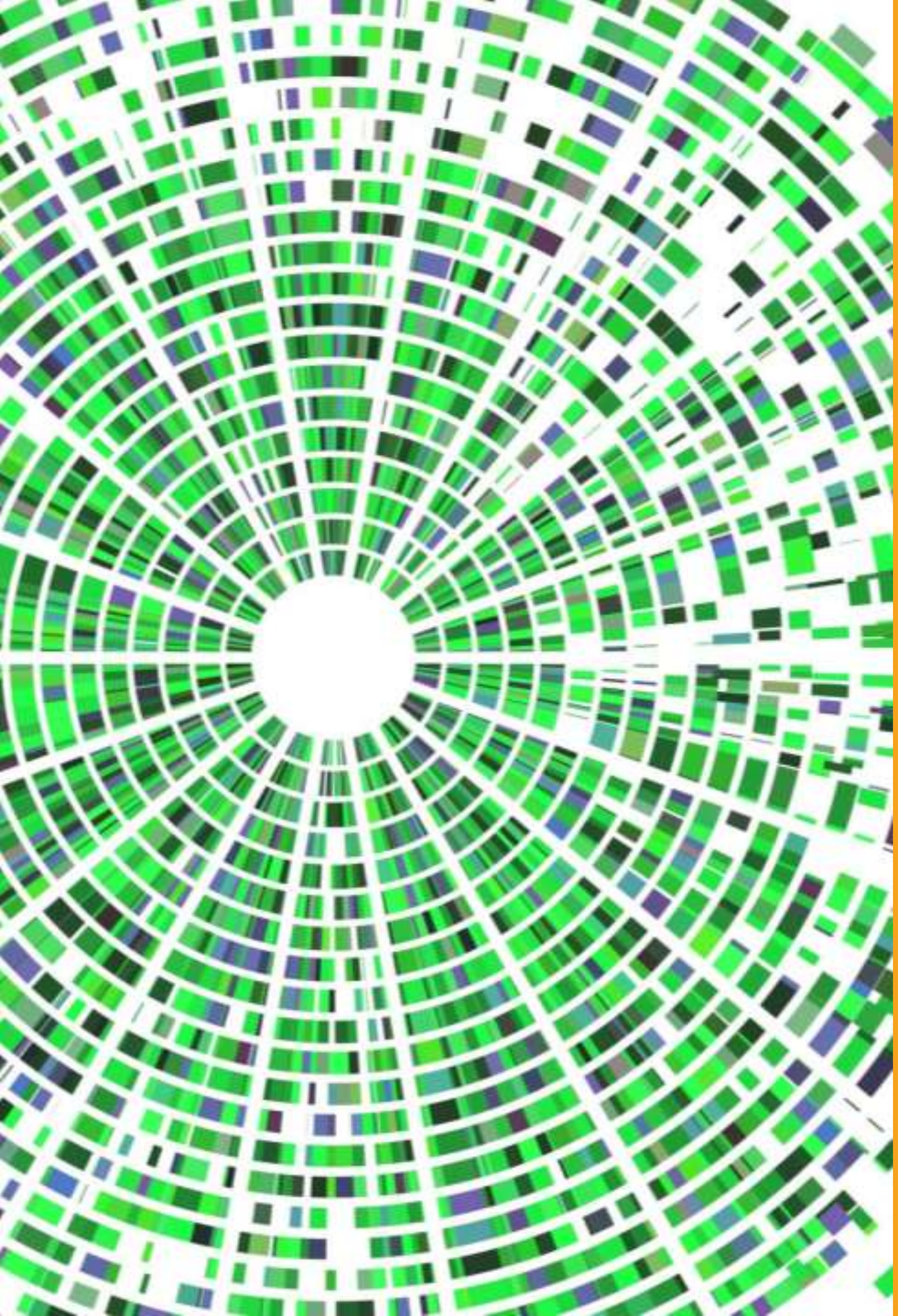

## **Exploration**

Allow you to understand your data better



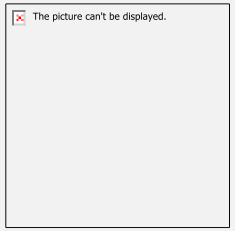

## **Communication**

Allow you to communicate what you want



# DATA VISUALISATION - DESIGN APPROACHES

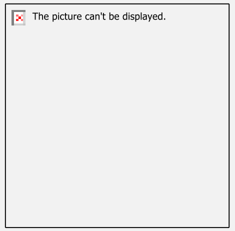
# COMMUNICATION - DESIGN




## Purpose

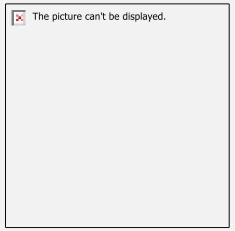
Who is the audience?

What story do we want to tell?




## Chart Types

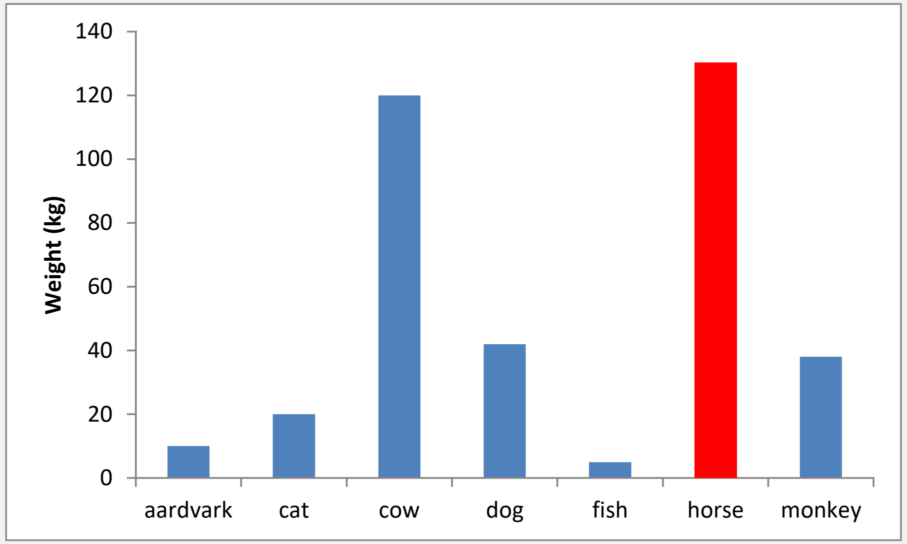
Different types for different purposes



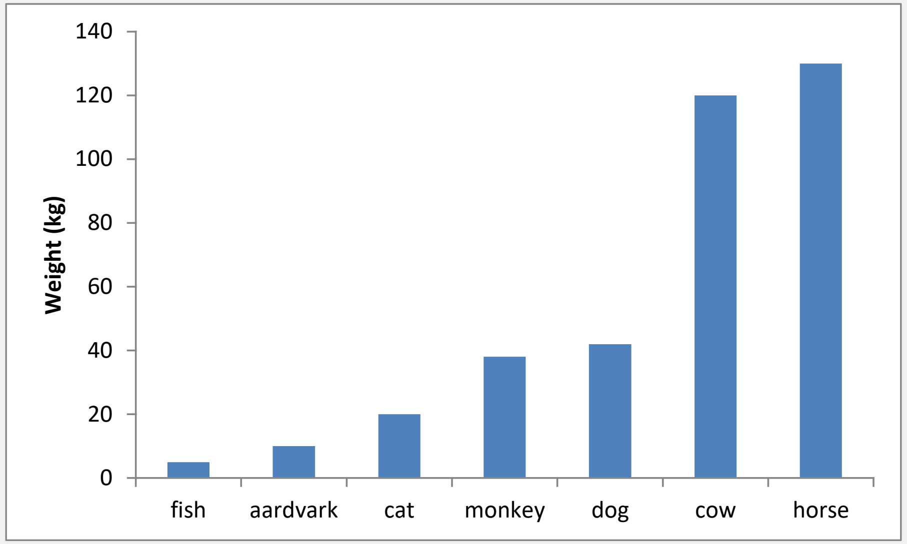

## Construction

Building blocks of plots

# PURPOSE

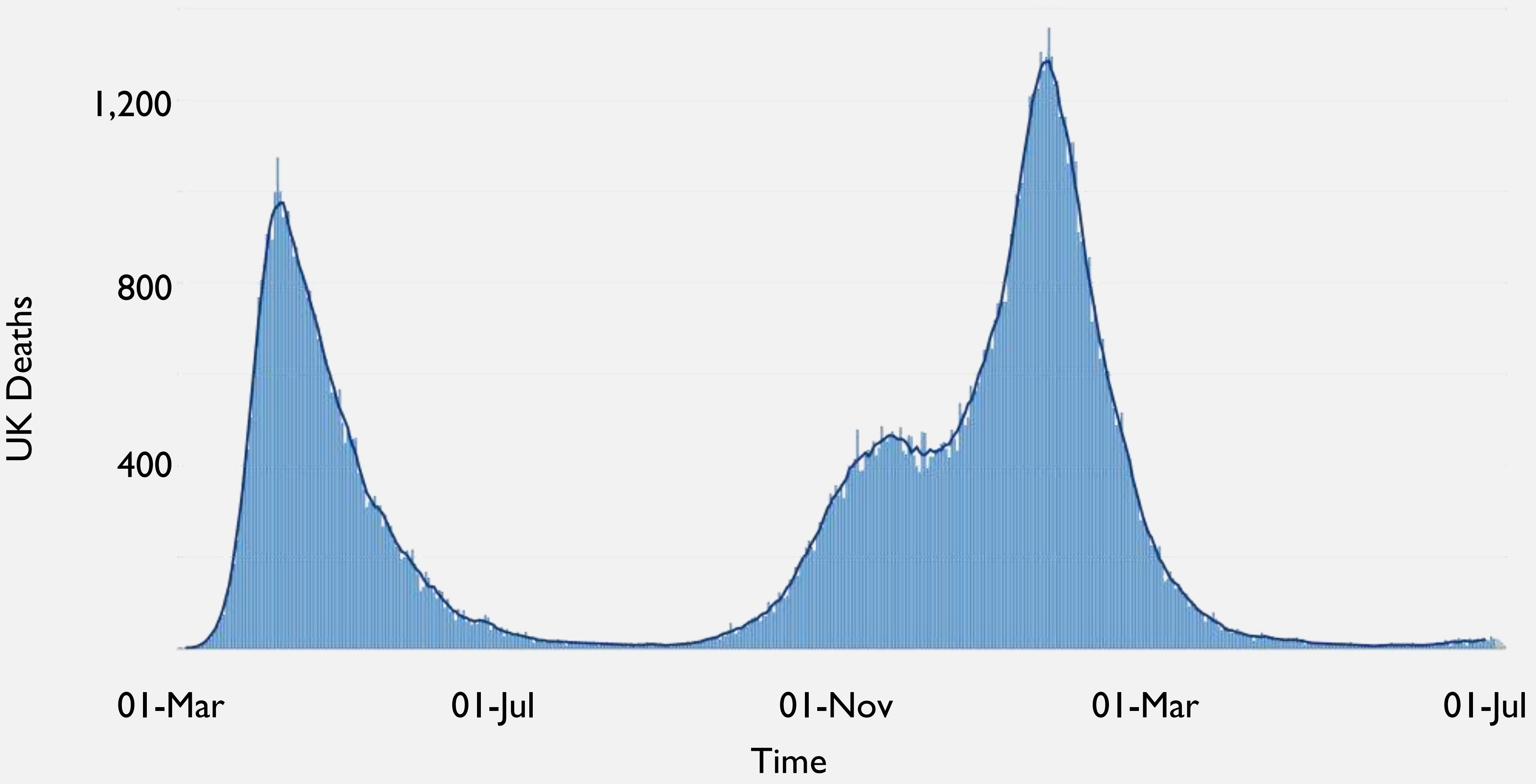


140
120
100
80
60
40
20
0
Weight (kg)
aardvark
cat
cow
dog
fish
horse
monkey

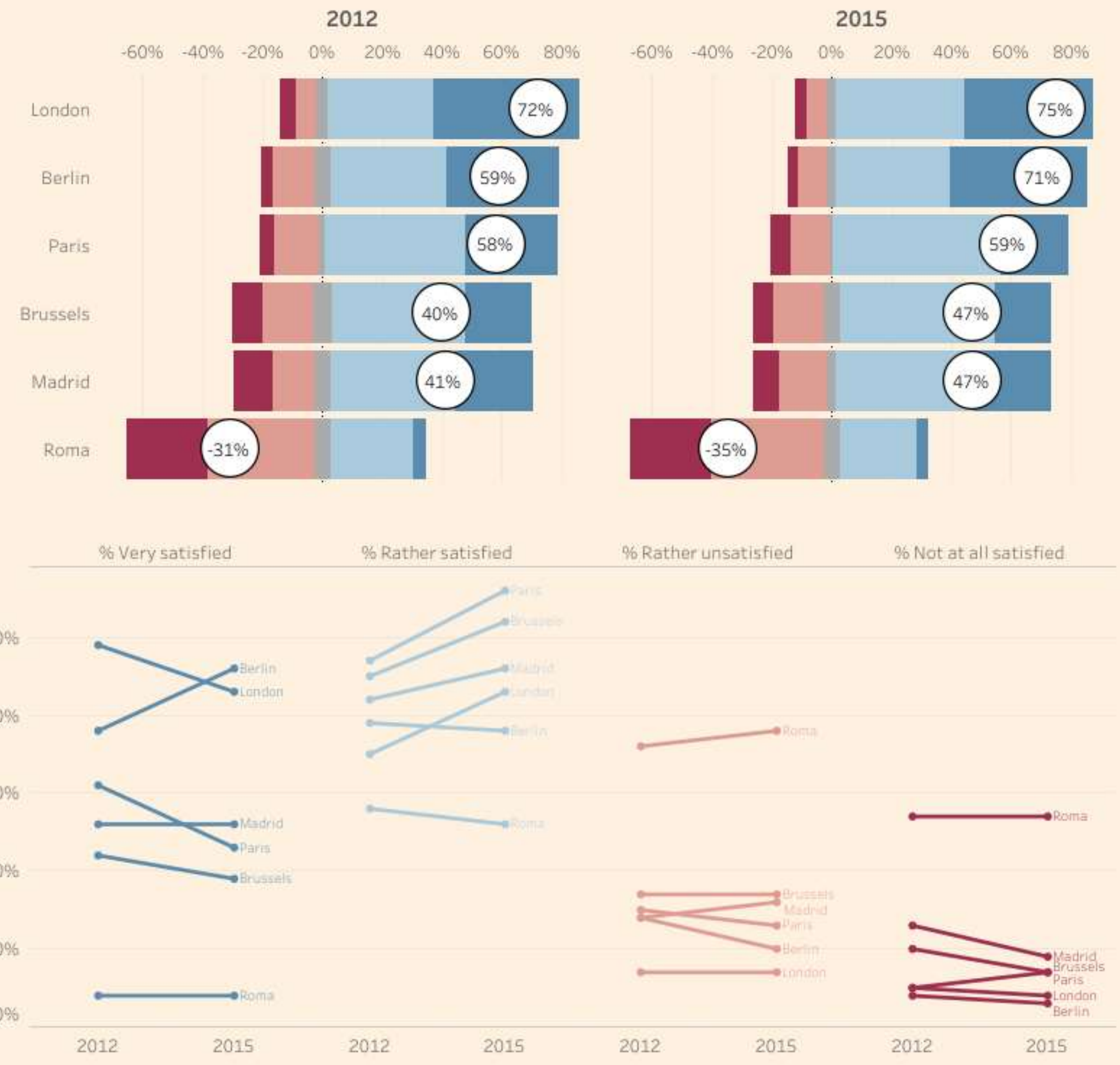
140
120
100
80
60
40
20
0
Weight (kg)
fish
aardvark
cat
monkey
dog
cow
horse

https://coronavirus.data.gov.uk/



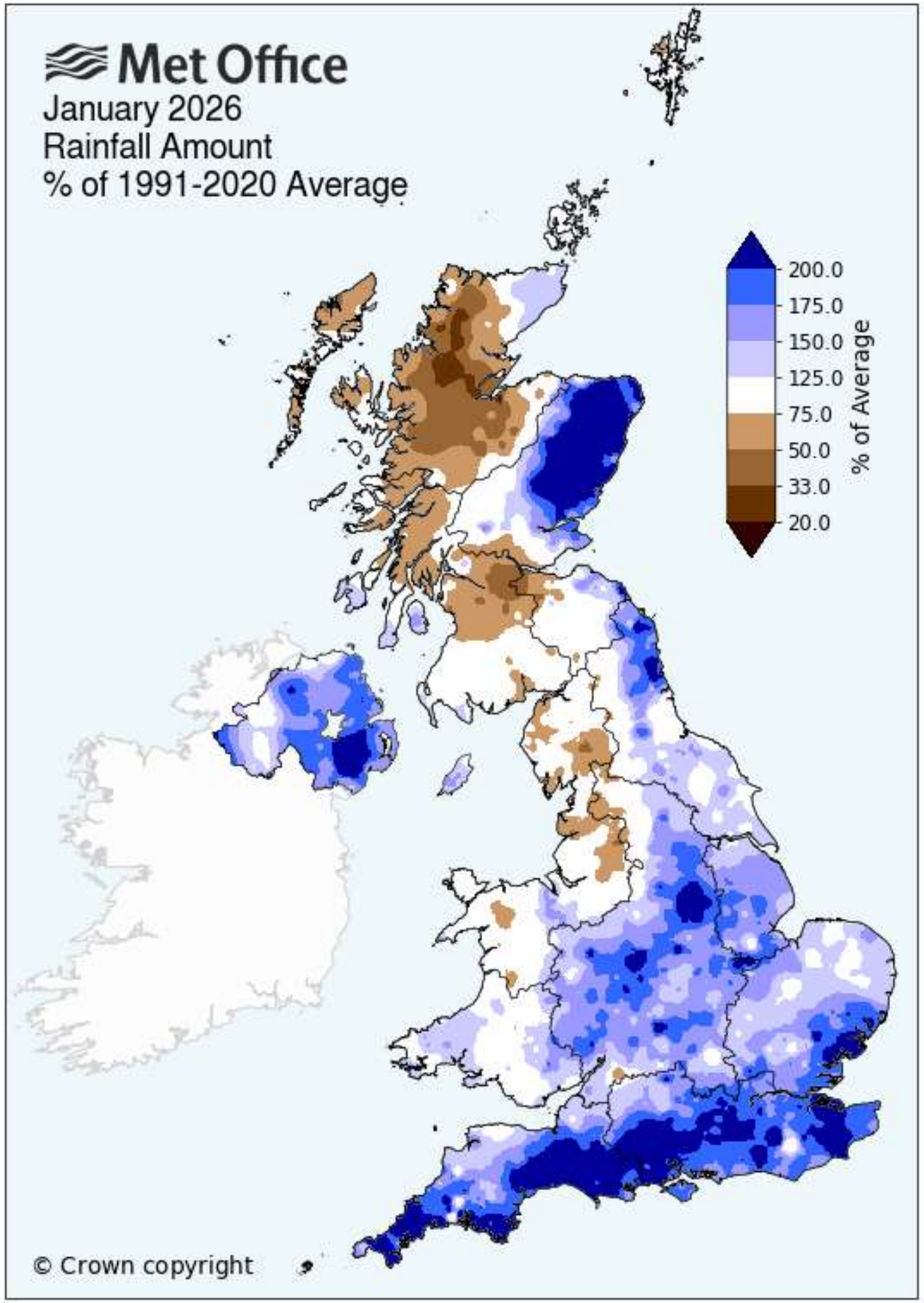
# How satisfied are people with public transportation in some of Europe's biggest cities?

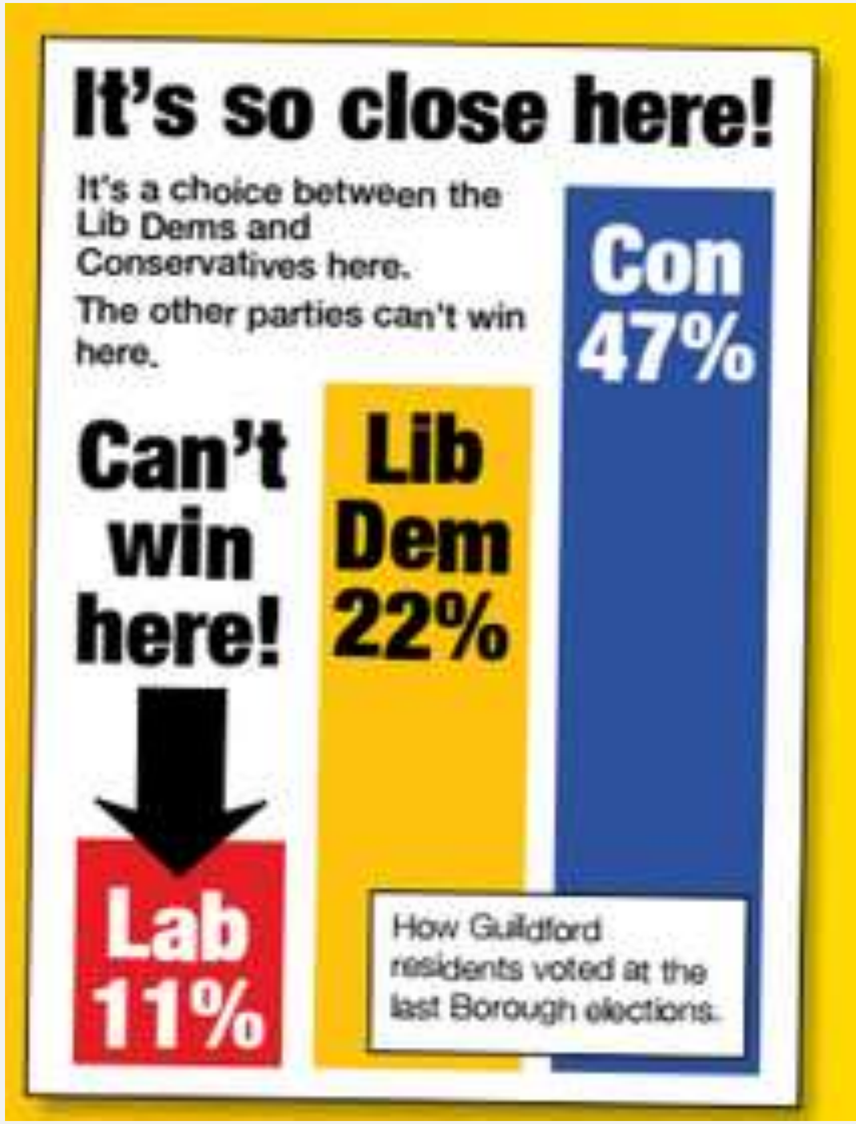
Overall satisfaction with public transport has increased in London, Berlin, Paris, Brussels and Madrid, while in Rome the vast majority is even more dissatisfied .




Source: Eurostat

Created by @VizWizBI

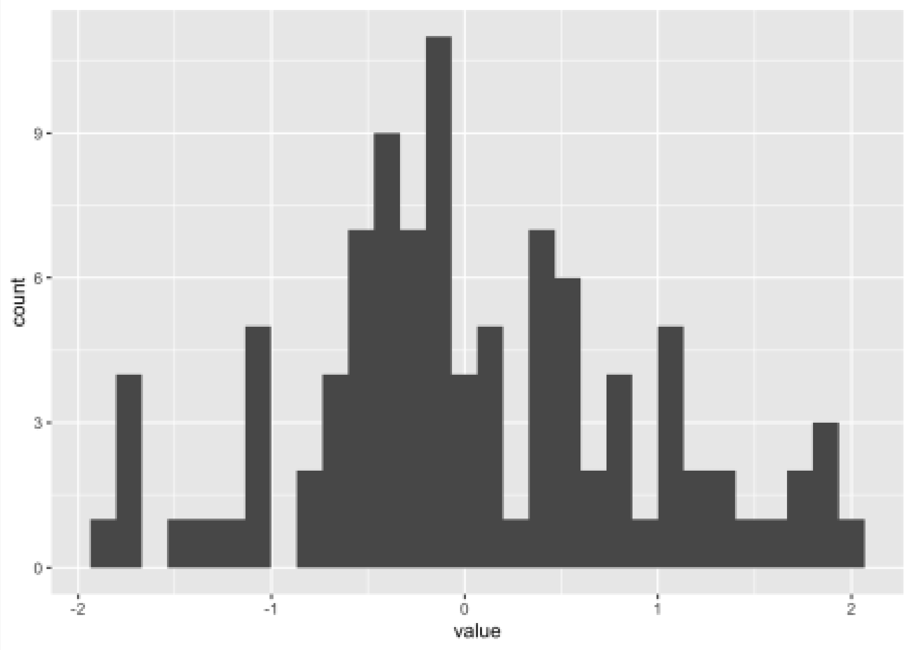

Met Office
January 2026
Rainfall Amount
% of 1991-2020 Average
200.0
175.0
150.0
125.0
75.0
50.0
33.0
20.0
% of Average
© Crown copyright


It's so close here!
It's a choice between the Lib Dems and Conservatives here.
The other parties can't win here.
Can't win here!
Lab 11%
Lib Dem 22%
Con 47%
How Guildford residents voted at the last Borough elections.

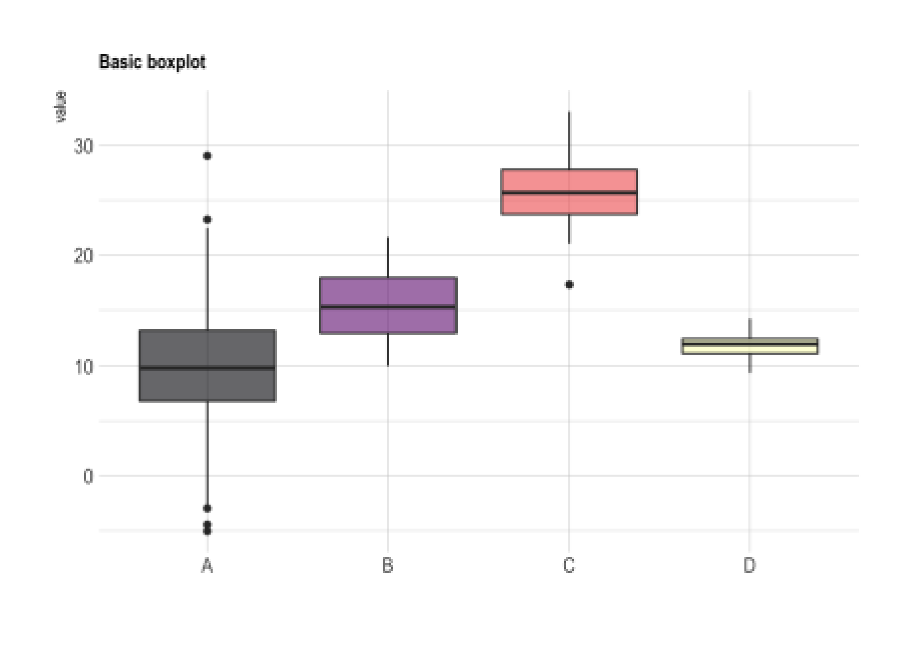
# CHART TYPES – CATEGORISATION & PURPOSE

- Distribution
- Correlation
- Ranking
- Composition
- Change over time
- Maps
- Networks and Flows

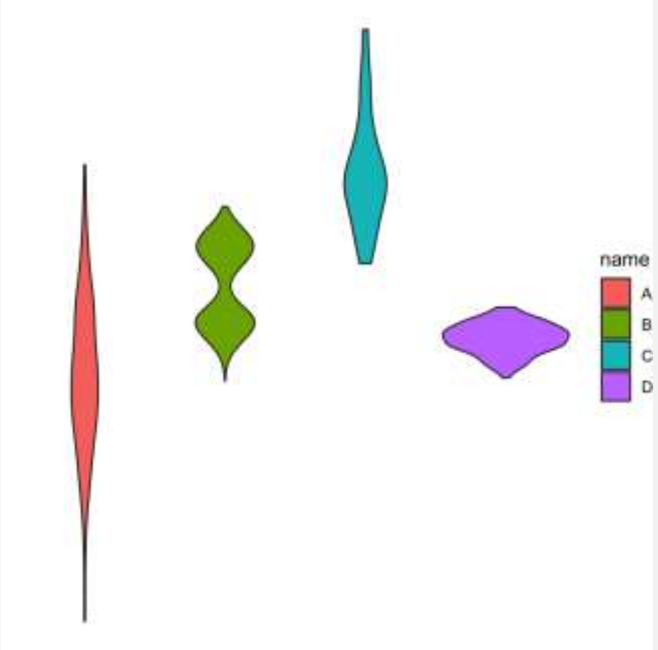
# DISTRIBUTION




Histogram

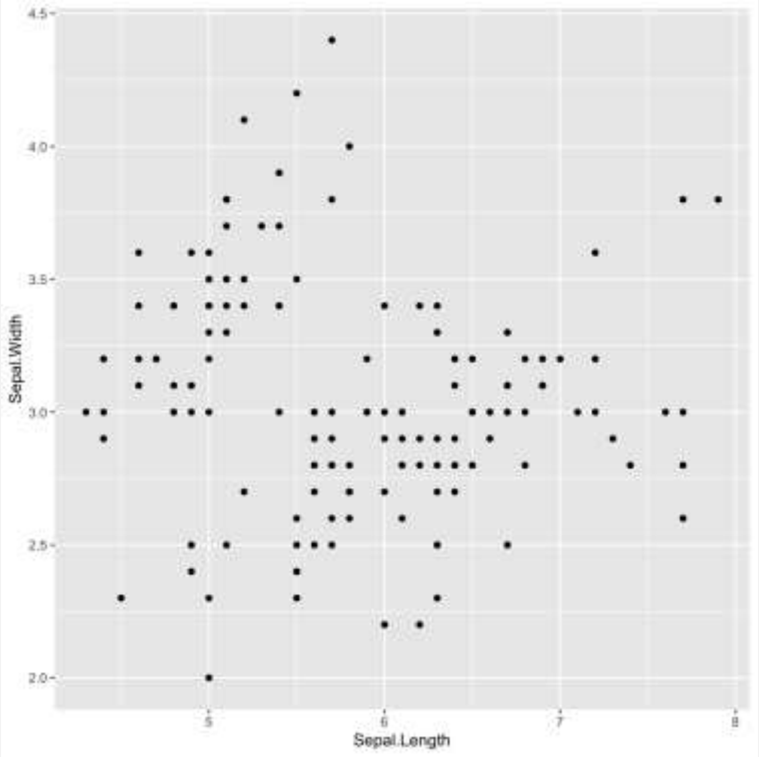



Boxplot

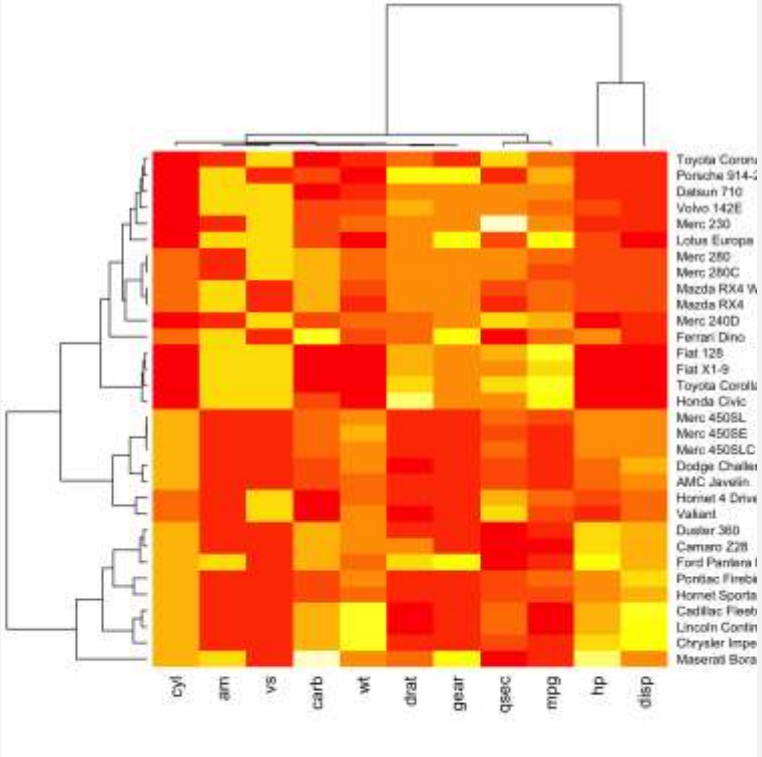



Violin

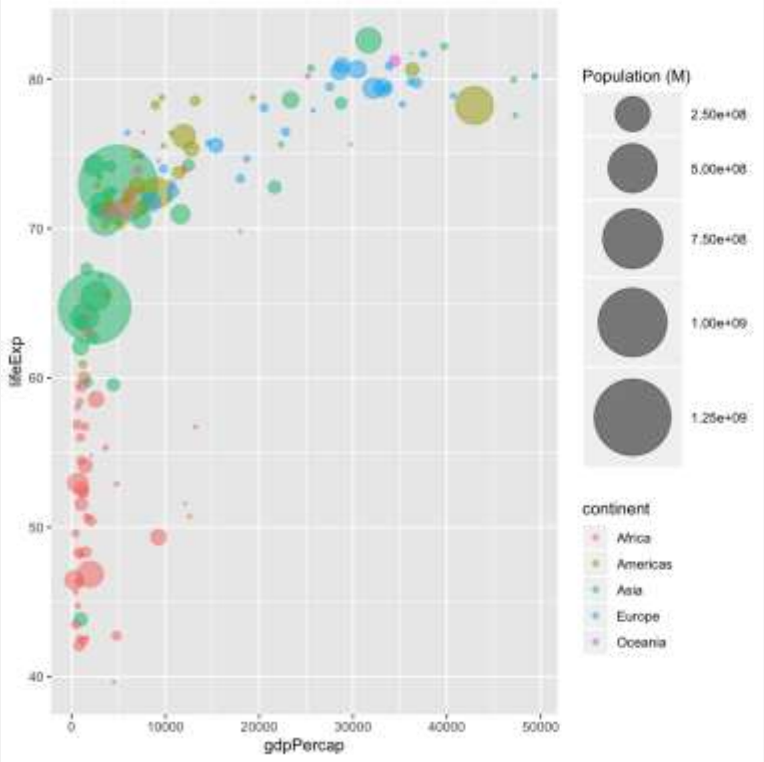
# CORRELATION




Scatterplot

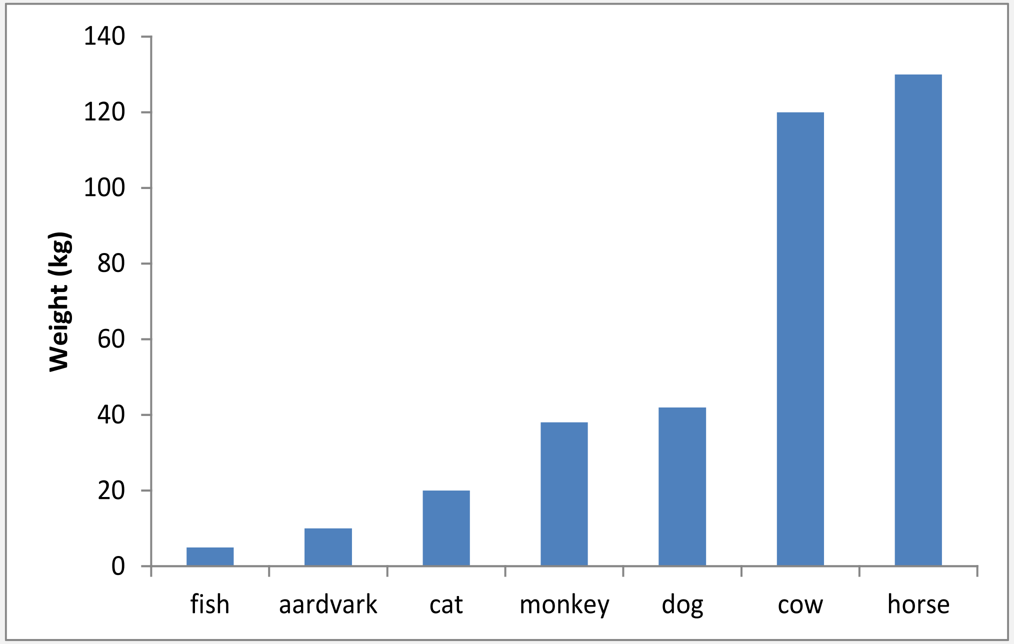



Heatmap

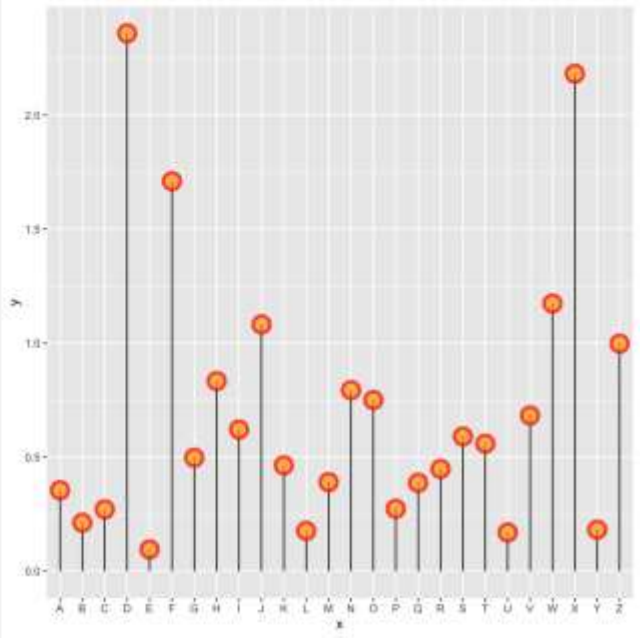



Bubble

# RANKING



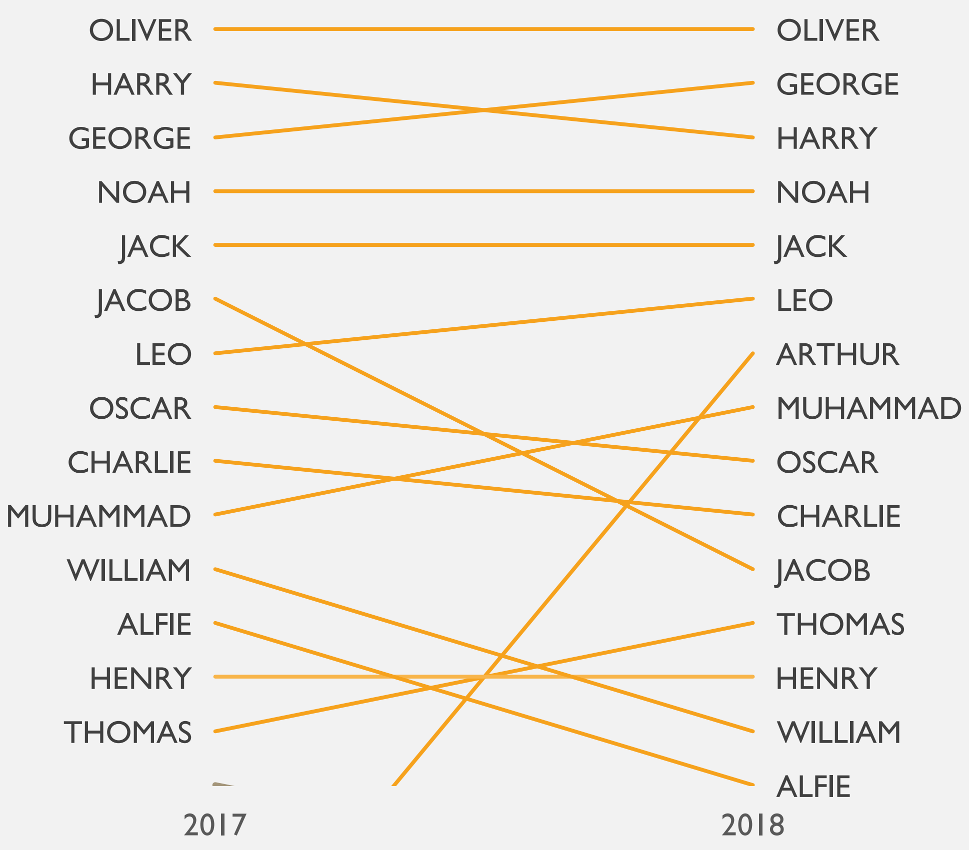

Bar

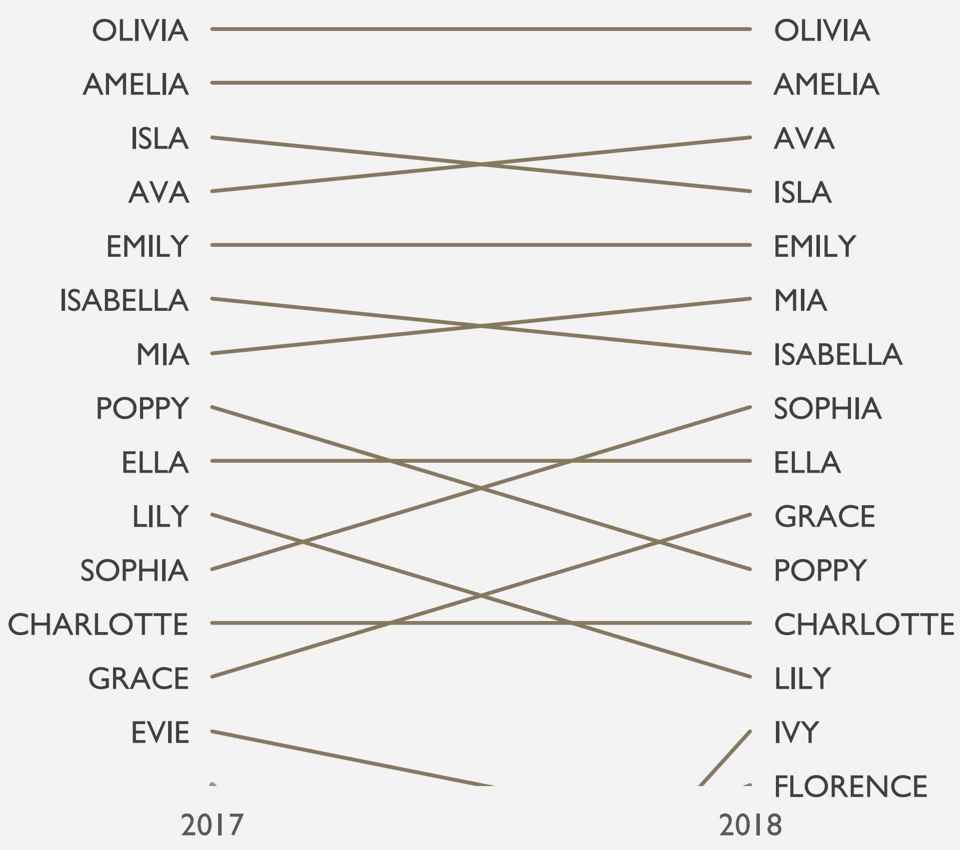


Lollipop


Spider

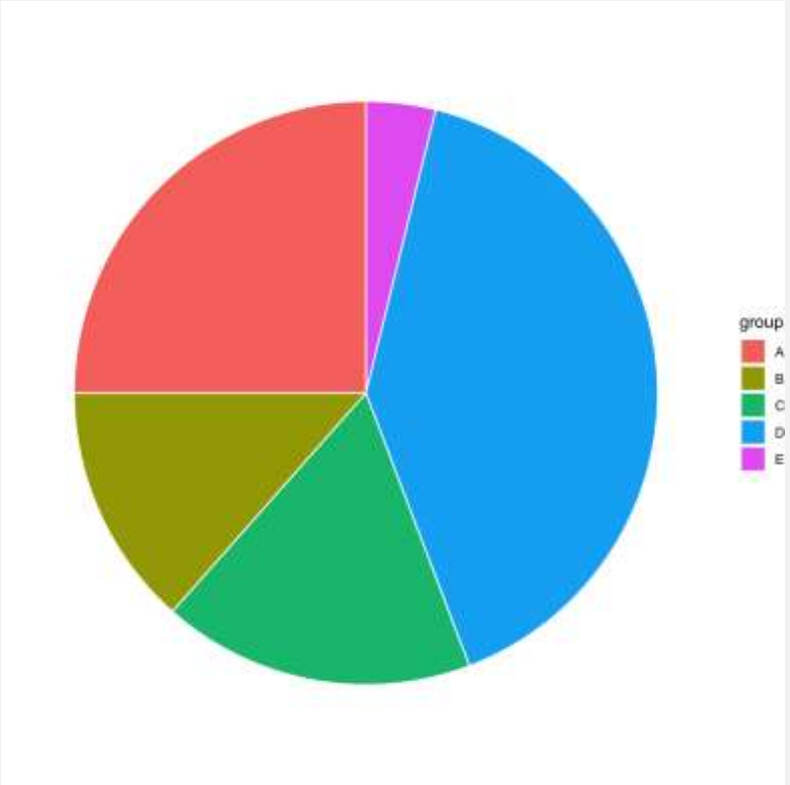
# RANKING



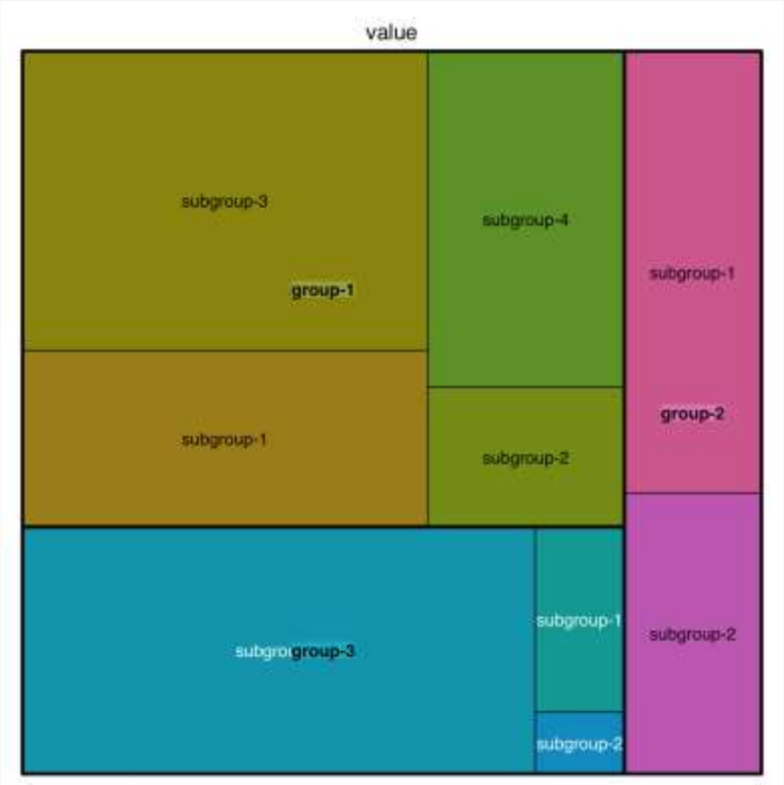




Parallel / Bumps

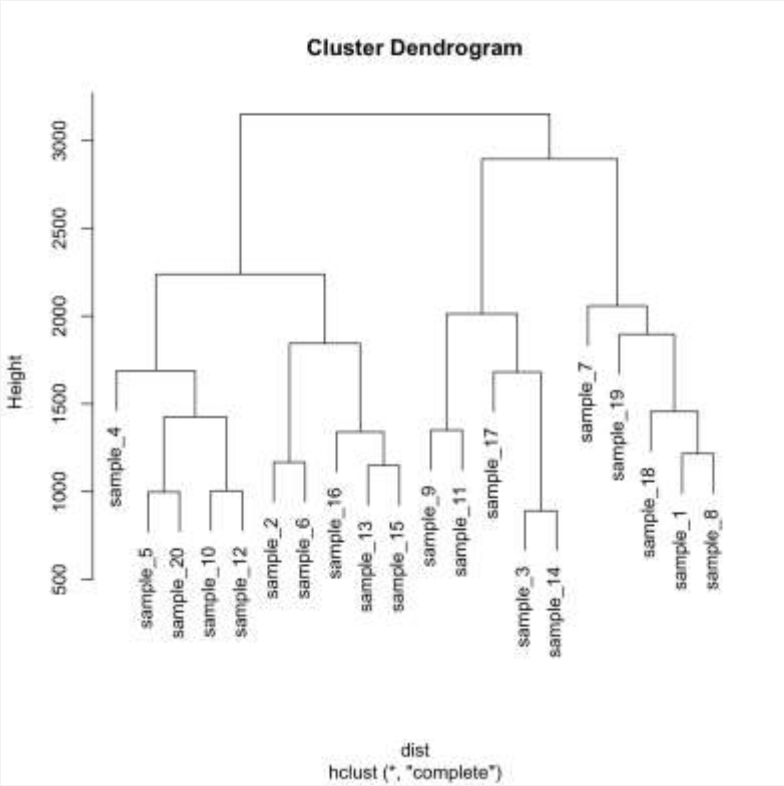
# COMPOSITION




Pie

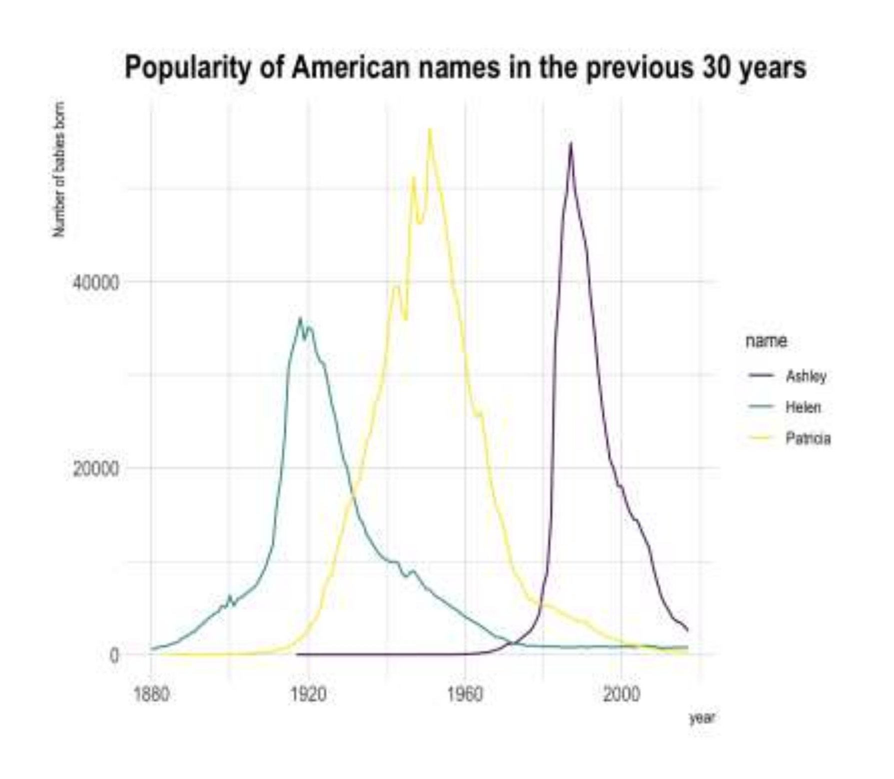


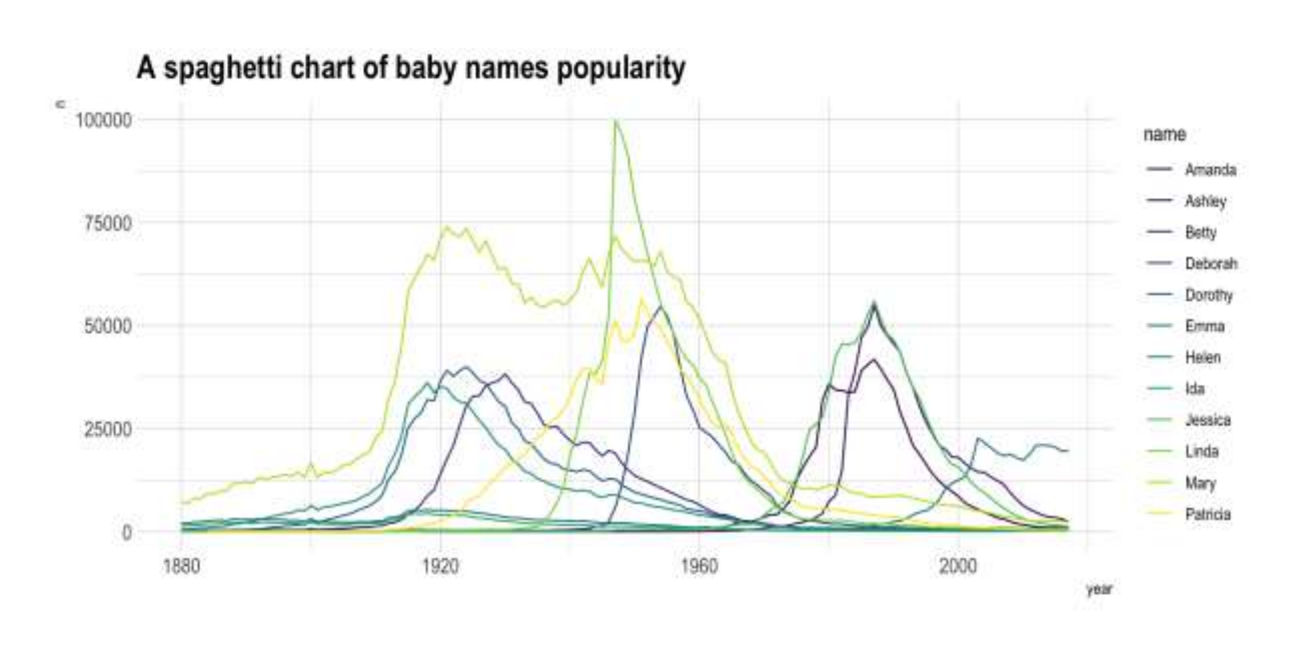

Treemap

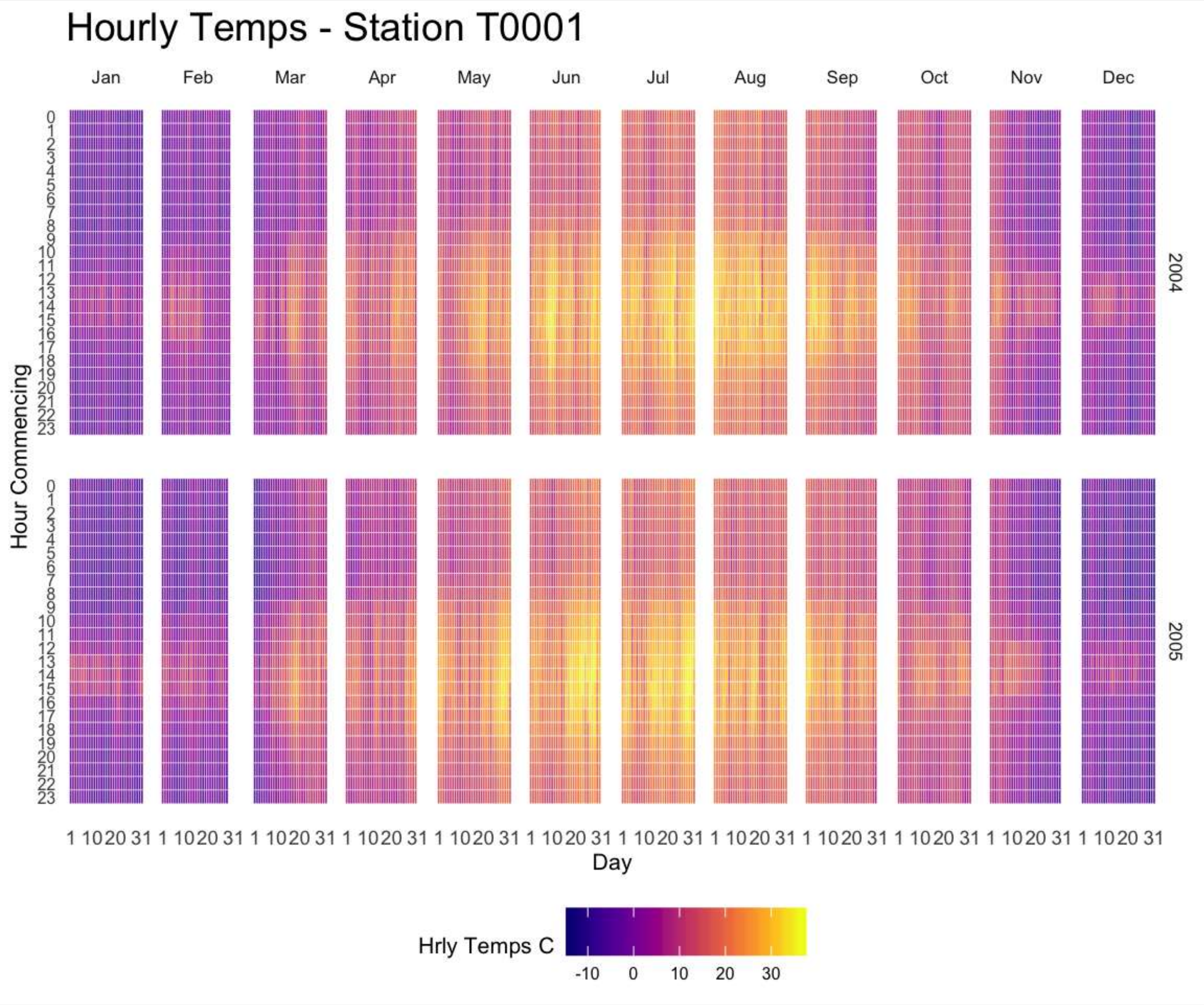



Dendrogram

# CHANGE OVER TIME

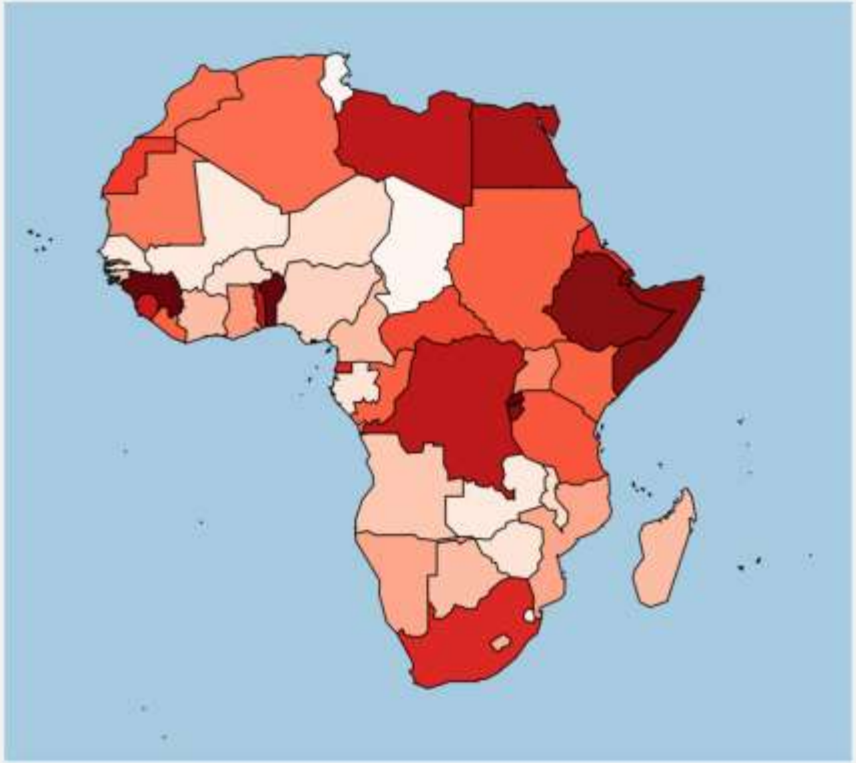


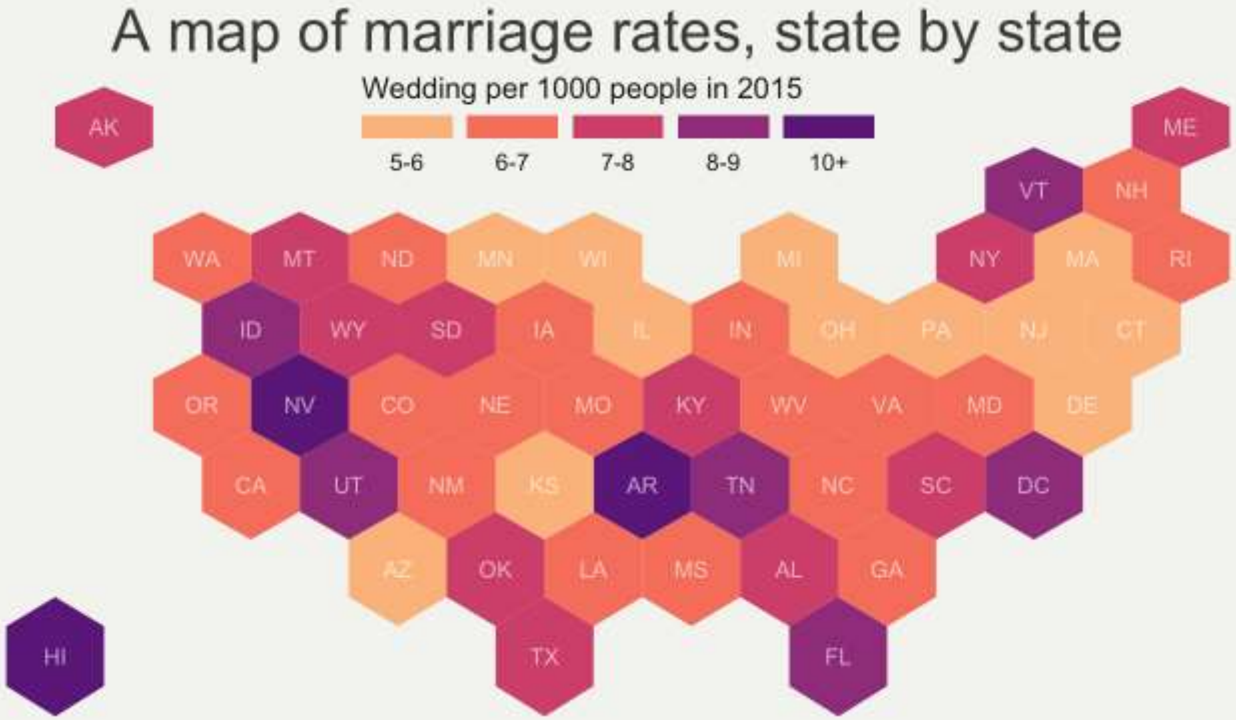


Hourly Temps - Station T0001
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Hour Commencing
2004
2005
Day
Hrly Temps C
-10 0 10 20 30

# MAPS



Choropleth




Hexbin

# CONSTRUCTION CONCEPTS

1. Marks and Channels
2. Colour
3. Discriminability
4. Popout
5. Space & Layout

# I. MARKS AND CHANNELS

## Marks

### Geometrical Primitives

- Points
- Lines
- Shapes

Used to represent data

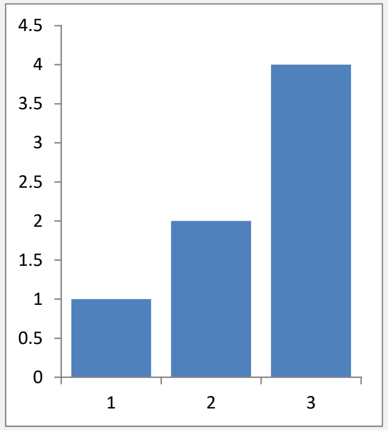
## Channels

### Appearance of a mark

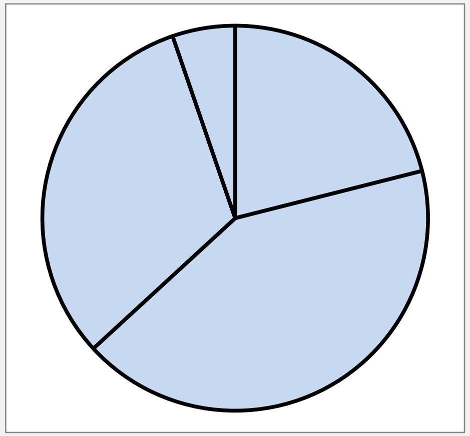
- Colour
- Length
- Position
- Size of Angle
- Area

Used to encode information

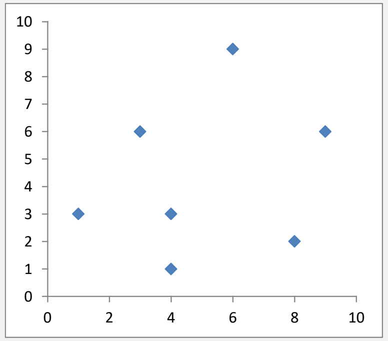
# FIGURES ARE A COMBINATION OF **MARKS** AND **CHANNELS**



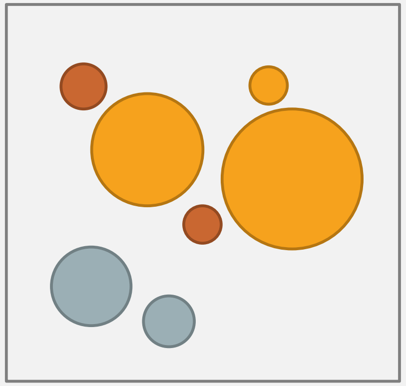

1 Mark:
	Rectangle
1 Channel:
	Length of longest side



1 Mark:
	Circle segment
1 Channel:
	Angle




1 Mark:
	Diamond shape
2 Channels:
	X position,
	Y position



1 Mark:
	Circle
4 Channels:
	X position
	Y position
	Area
	Colour

# EXPRESSIVENESS
# MATCHING DATA & CHANNELS

## Types of Data

### Quantitative

- Height, Length, Weight, Expression etc.

### Ordered

- Small, Medium, Large
- January, February, March

### Categorical

- WT, Mutant1, Mutant2
- GeneA, GeneB, GeneC

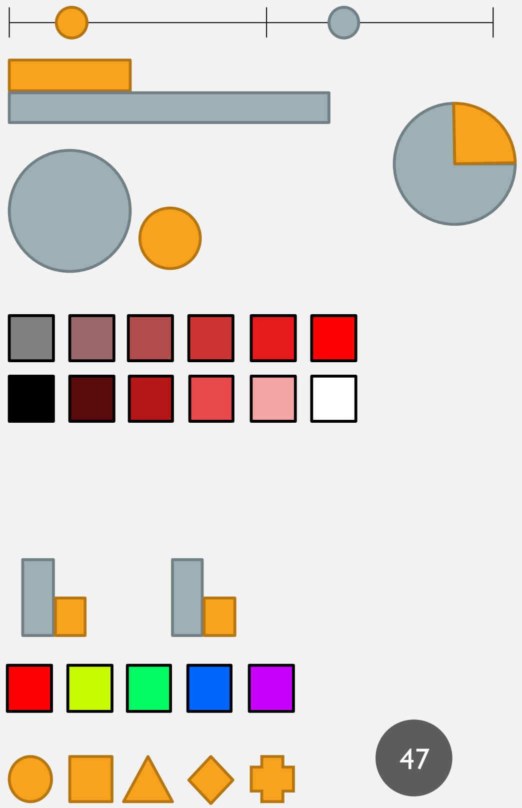
## Types of Channel

### Quantitative

- Position on scale
- Length
- Angle
- Area
- Colour (saturation)
- Colour (luminosity)

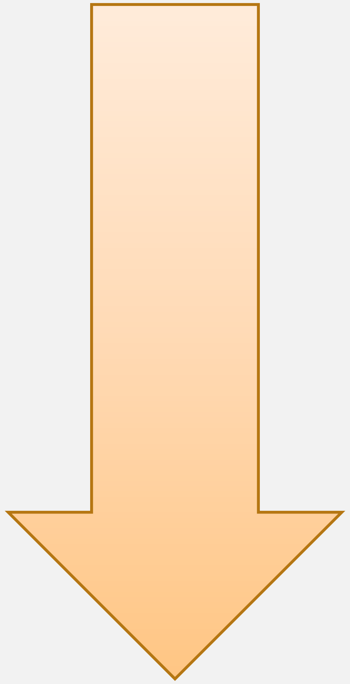
### Qualitative

- Spatial Grouping
- Colour (hue)
- Shape



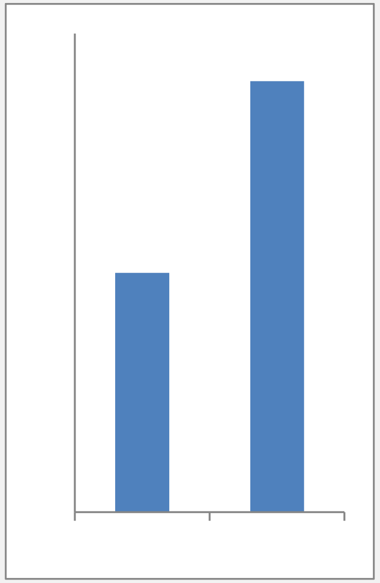
# EFFECTIVENESS OF QUANTITATIVE REPRESENTATIONS

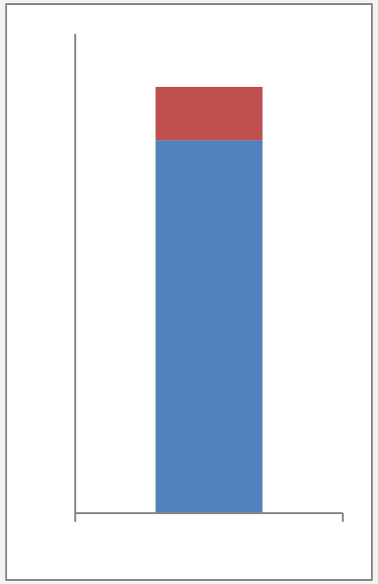
Good quantitation

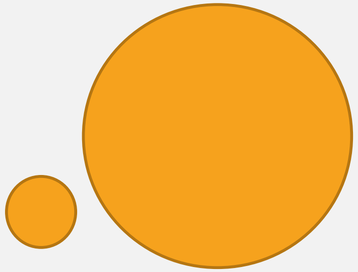


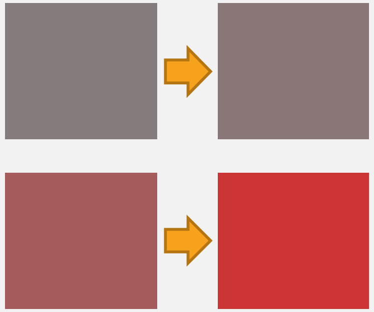
Poor quantitation

- Bar chart (lengths)
- Stacked bar chart (lengths)
- Pie charts (angle)
- Histograms (rectangular area)
- Bubble plots (circular area)
- Colour (luminance)
- Colour (saturation)





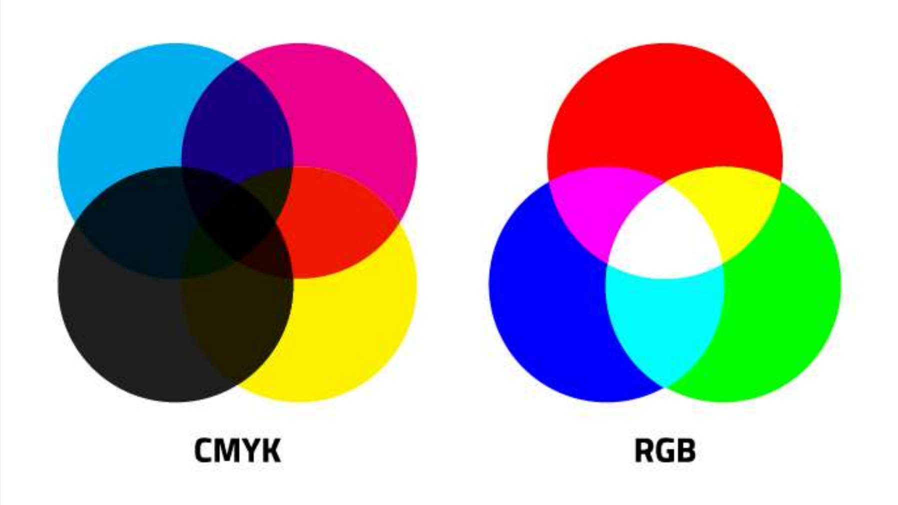




# 2. COLOUR

## Technical representations

- Red + Green + Blue (RGB)
- Cyan + Magenta + Yellow + Black (CMYK)




Printing Computers

## Perceptual representation

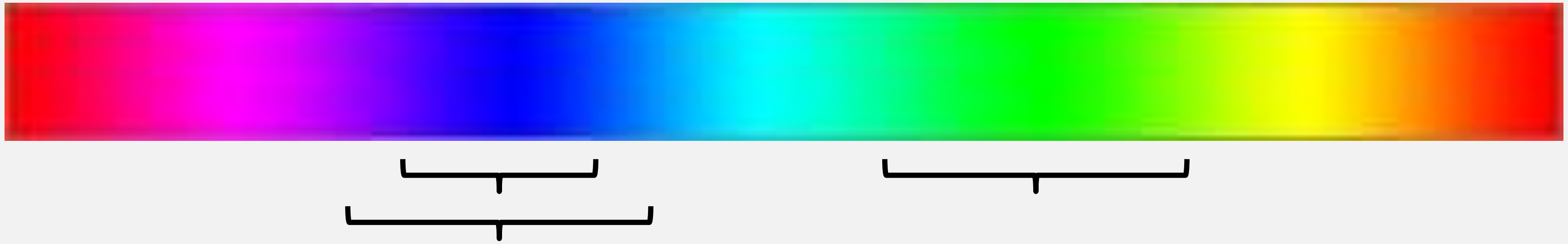
Hue + Saturation + Luminance (HSL)

# HSL REPRESENTATION

- Hue = Shade of colour = Qualitative
- Saturation = Amount of colour = Quantitative
- Luminance = Amount of white = Quantitative

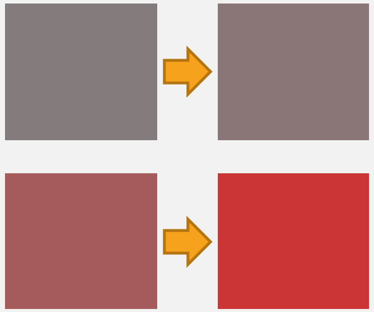
Humans have no innate quantitative perception of hue but we have learned some (cold – hot, rainbow etc.)

Our perception of hue is not linear

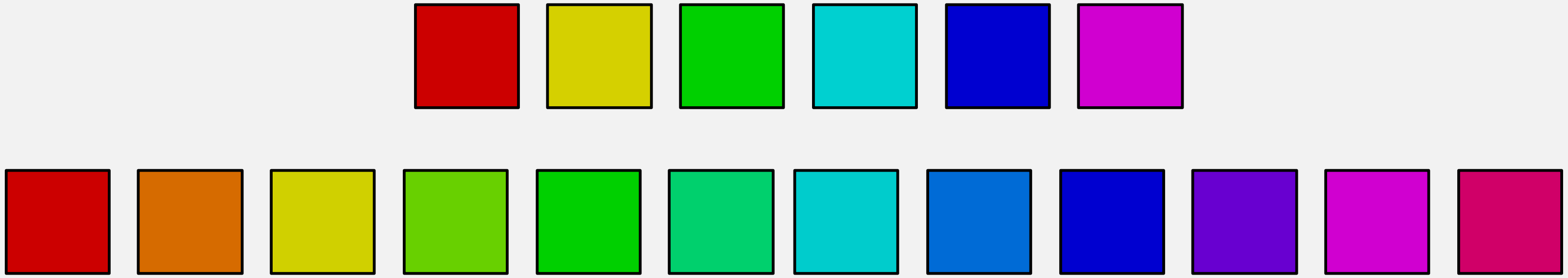


# COLOUR CHANNELS

Poor for quantitative data

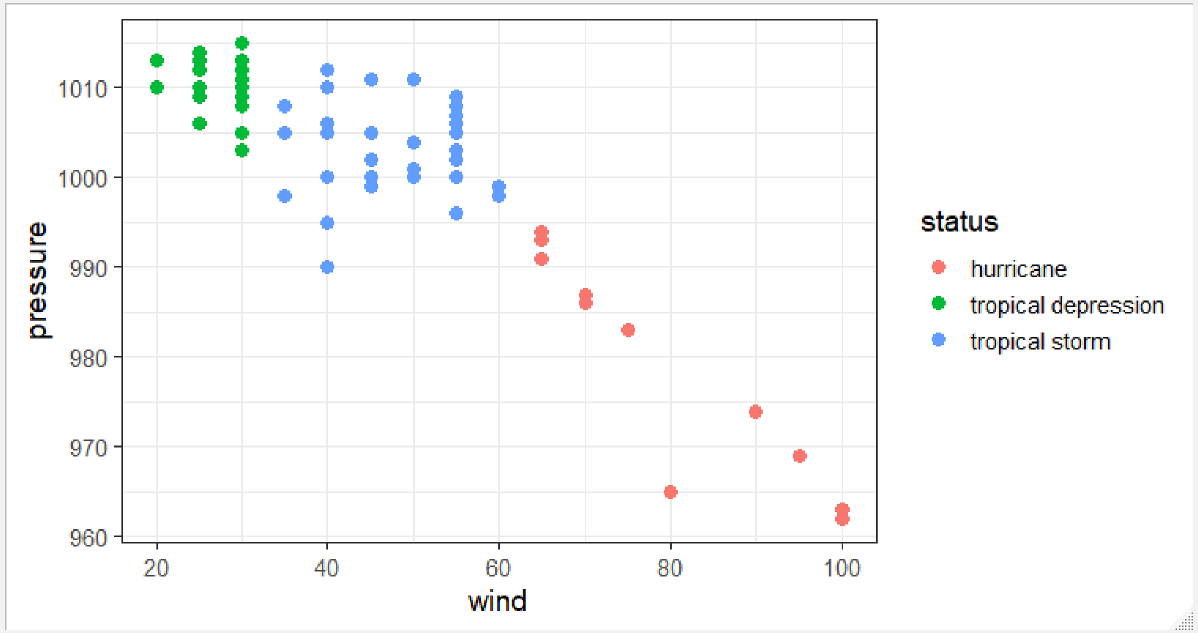


Better for qualitative

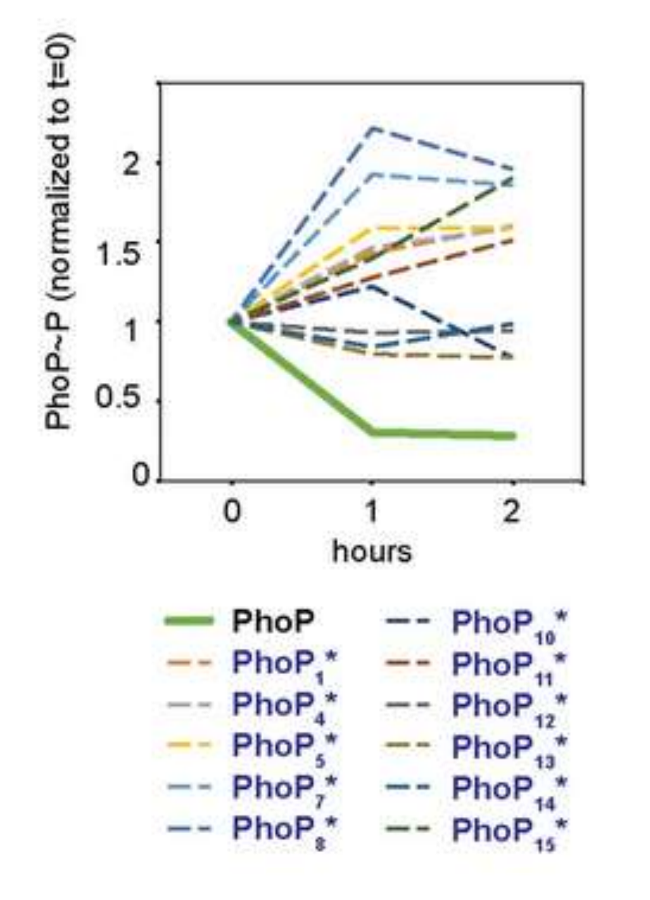


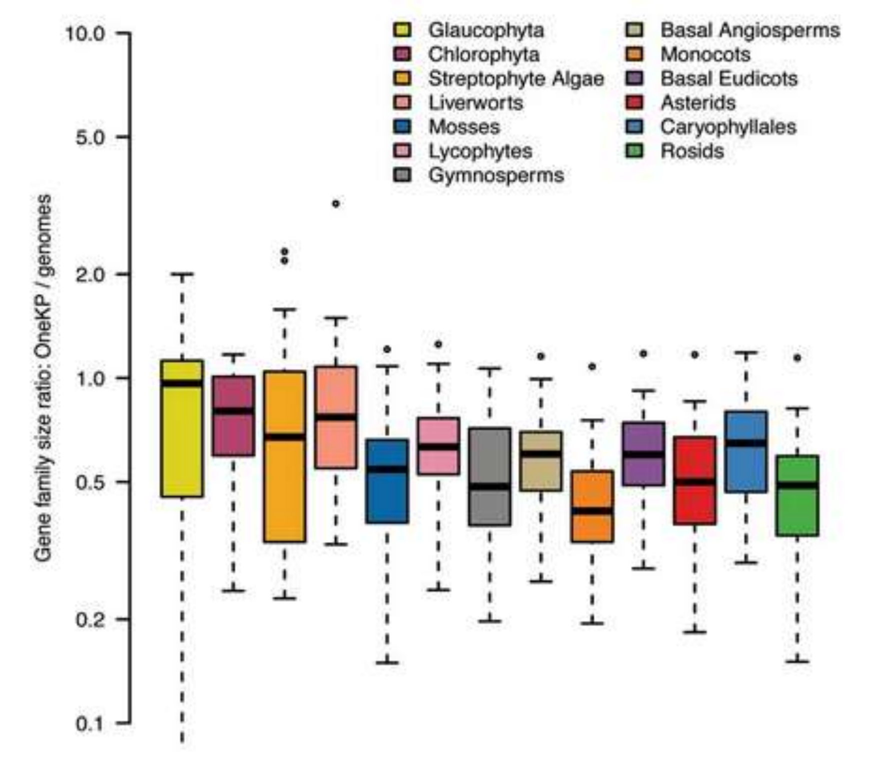
# 3. CATEGORICAL DATA - DISCRIMINABILITY

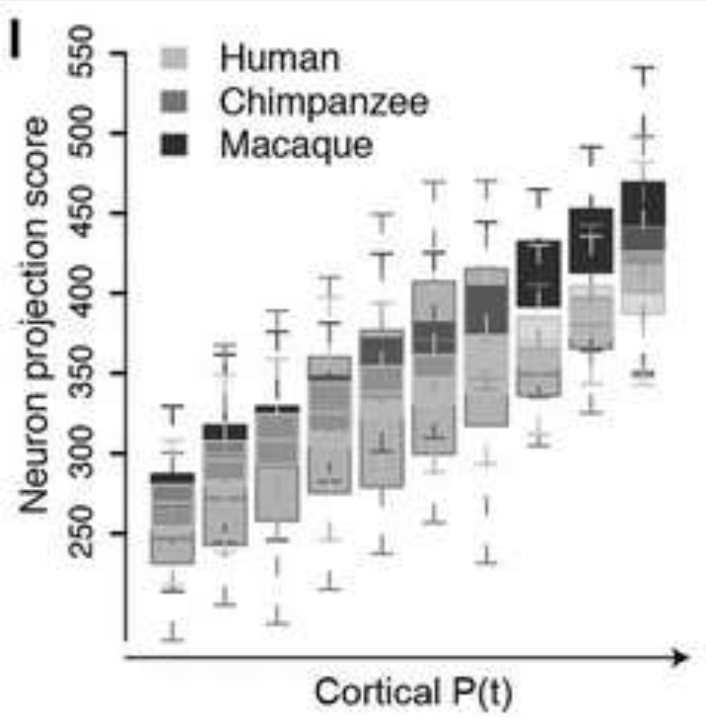
- If you encode categorical data, are the differences between categories easy for the user to perceive correctly?



# COLOUR DISCRIMINATION



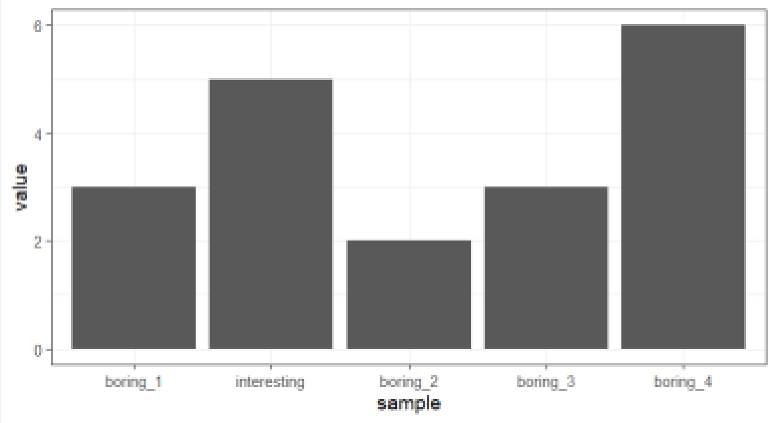



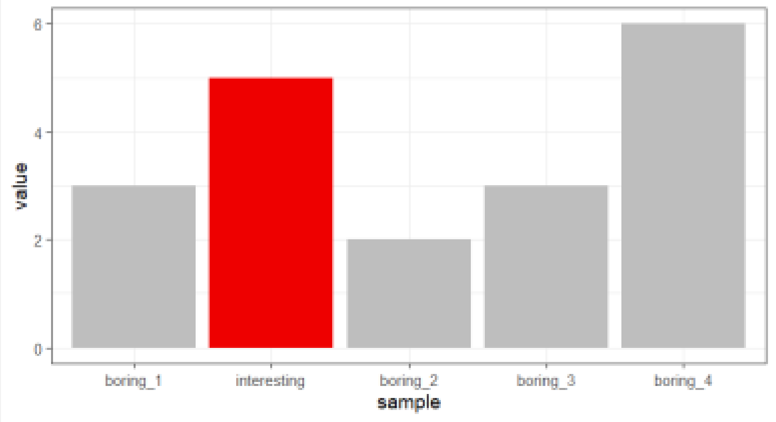
# QUALITATIVE DISCRIMINATION

- How many (fillable) shapes can you discriminate?

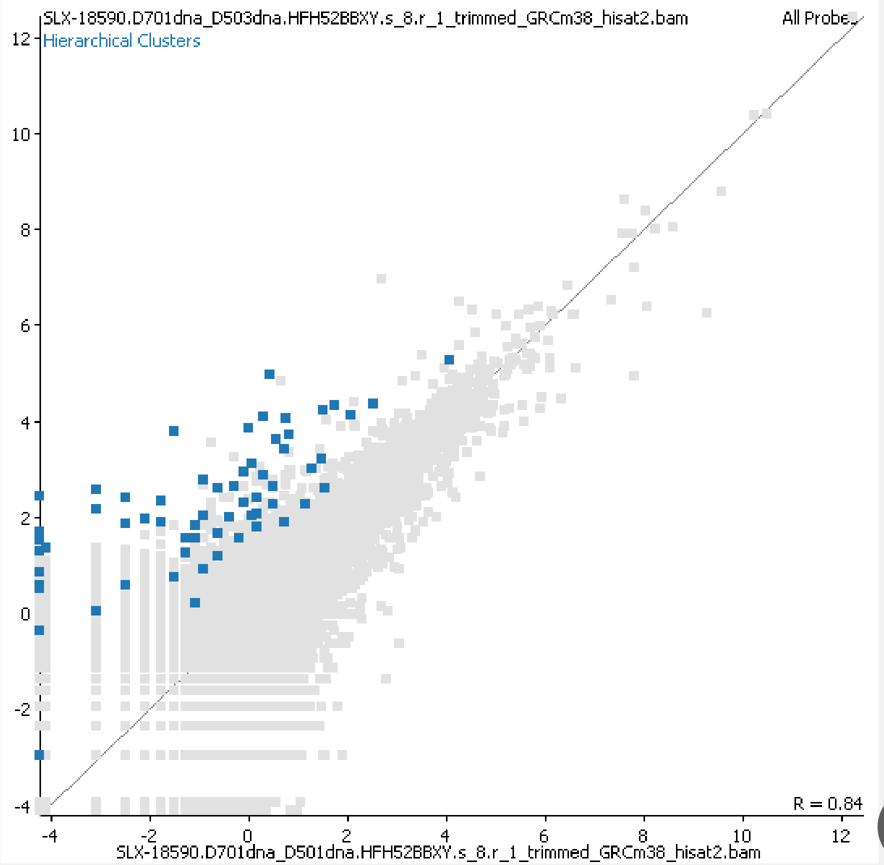
# 4. POPOUT



value
sample
boring_1
interesting
boring_2
boring_3
boring_4

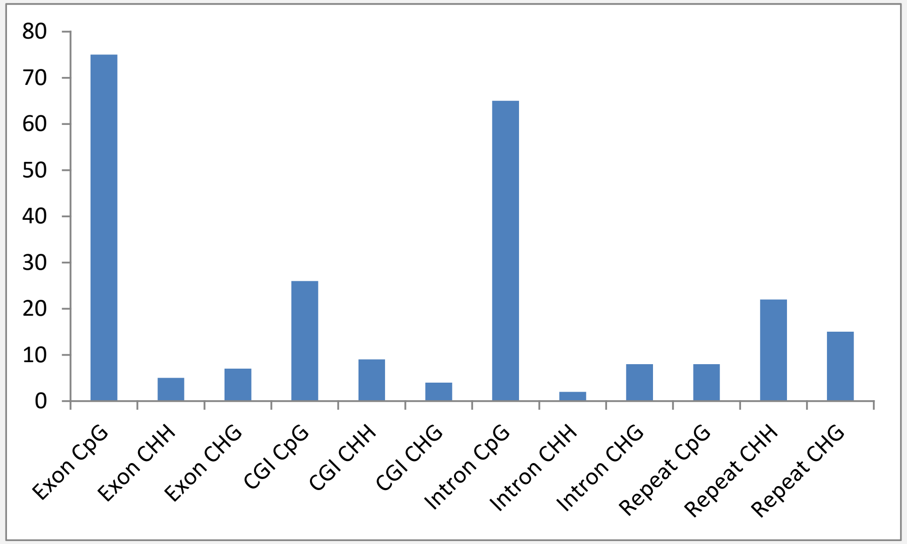



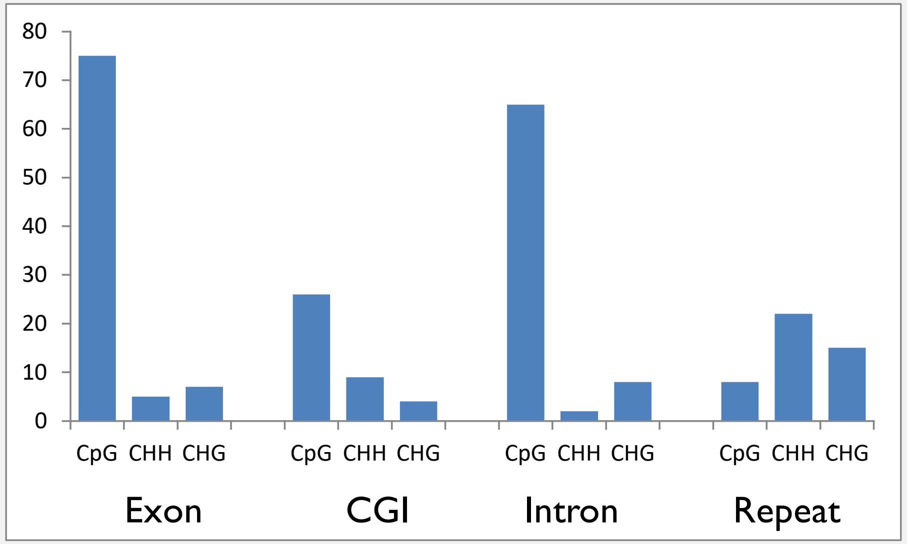
value
sample
boring_1
interesting
boring_2
boring_3
boring_4

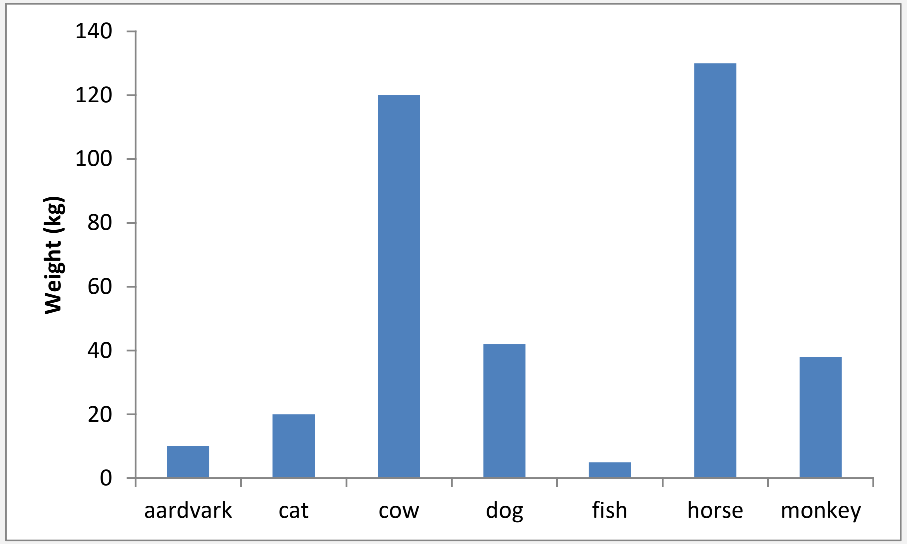
SLX-18590.D701dna_D503dna.HFH52BBXY.s_8.r_1_trimmed_GRCm38_hisat2.bam
All Probes
Hierarchical Clusters
R = 0.84
SLX-18590.D701dna_D501dna.HFH52BBXY.s_8.r_1_trimmed_GRCm38_hisat2.bam

# 5. SPACE & LAYOUT

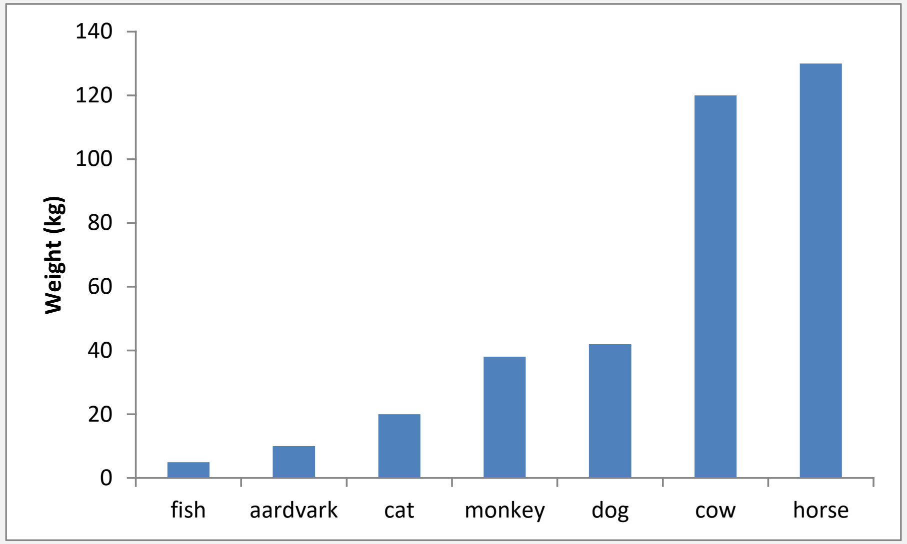


# ORDERING



Weight (kg)
140
120
100
80
60
40
20
0
aardvark
cat
cow
dog
fish
horse
monkey

Weight (kg)
140
120
100
80
60
40
20
0
fish
aardvark
cat
monkey
dog
cow
horse

# COMMUNICATION - DESIGN

The picture can't be displayed.

## Purpose

Who is the audience?

What story do we want to tell?

The picture can't be displayed.

## Chart Types

Different types for different purposes

The picture can't be displayed.

## Construction

Building blocks of plots

# EXERCISE

- Split into groups of 4
- Find a visualisation online to share with the group
- Send the link to my email (mdc31@cam.ac.uk)

Be prepared to collectively comment on

- Purpose, audience and effectiveness
- Specific design choices:
  - Marks and channels
  - Colours
  - Discrimination
  - Separability
  - Space & layout